



# CMPT 413/713: Natural Language Processing

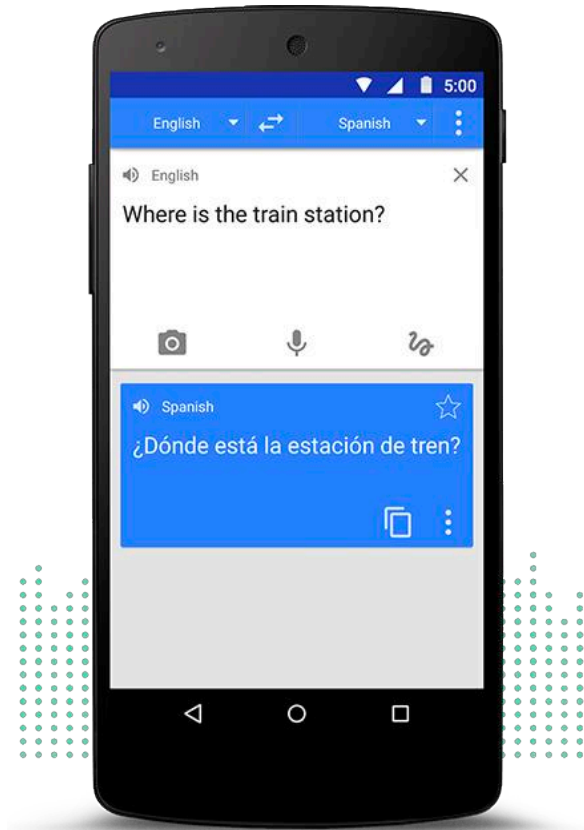
Angel Xuan Chang

2025-01-06

# Today

- Intro to NLP
  - What is language? NLP vs computational linguistics.
  - Brief history of NLP
  - Levels of representation in language
  - Key questions of NLP
- Course Logistics

# NLP is everywhere



Google translate



Virtual assistants

# Information finding



when were the first pyramids built



All

Images

News

Shopping

Videos

More

Settings

Tools

About 20,300,000 results (0.67 seconds)

## 2630 BC

Most were built as tombs for the country's pharaohs and their consorts during the Old and Middle Kingdom periods. The earliest known Egyptian pyramids are found at Saqqara, northwest of Memphis. The earliest among these is the Pyramid of Djoser (constructed **2630 BC**–2611 BC) which was built during the third dynasty.

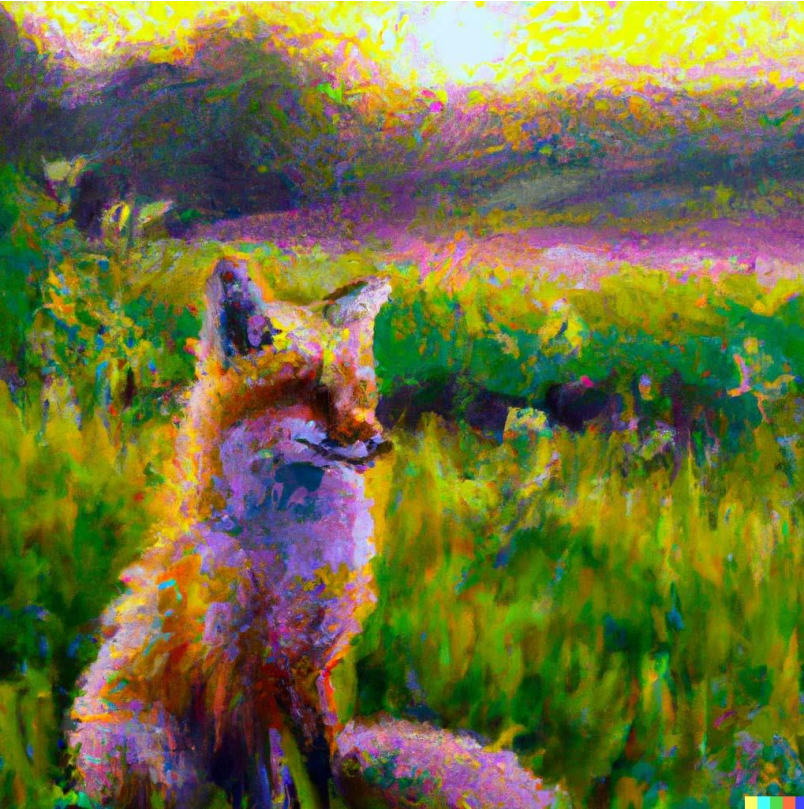
[The First Pyramids Built - Timeline Index](#)

[www.timelineindex.com](http://www.timelineindex.com) › [content](#) › [view](#)

# Text to image generation

 **OpenAI** DALL-E 2

a painting of a fox sitting in a field at sunrise in the style of Claude Monet



<https://openai.com/dall-e-2/>

 **Imagen**

A cute sloth holding a small treasure chest.  
A bright golden glow is coming from the chest.

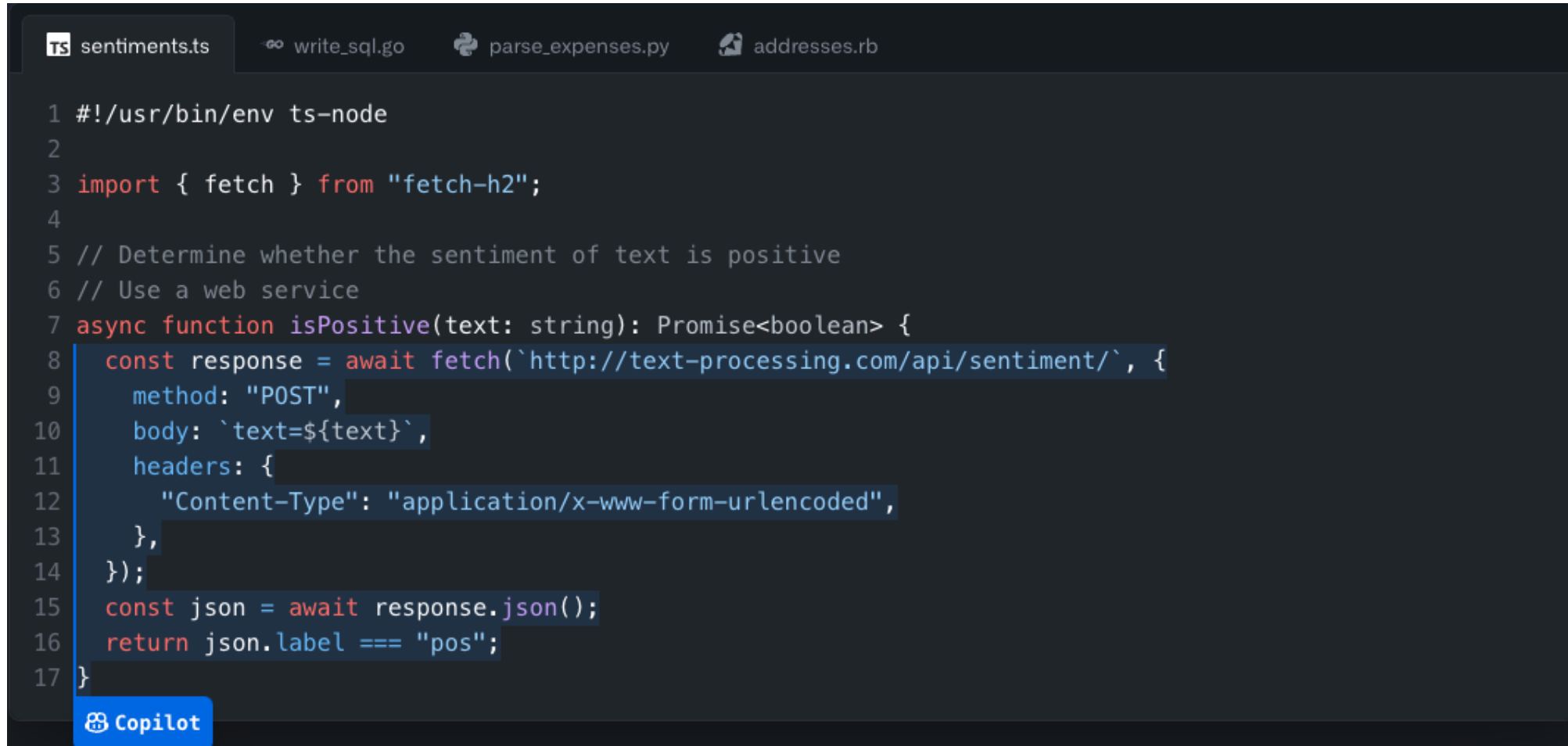


<https://imagen.research.google/>

Try it out yourself: <https://huggingface.co/spaces/stabilityai/stable-diffusion>,  
<https://www.craiyon.com/>, <https://www.midjourney.com/home/>

# Copilot

Generate code from text!



The screenshot shows a code editor with a dark theme. At the top, there are four tabs: 'sentiments.ts' (active), 'write\_sql.go', 'parse\_expenses.py', and 'addresses.rb'. The main editor area displays TypeScript code for a function named 'isPositive'. The code uses the 'fetch' API to send a POST request to 'http://text-processing.com/api/sentiment/'. The request body is a JSON object containing the text to be analyzed. The response is parsed as JSON, and the 'label' property is checked to see if it is 'pos'. The code is as follows:

```
1 #!/usr/bin/env ts-node
2
3 import { fetch } from "fetch-h2";
4
5 // Determine whether the sentiment of text is positive
6 // Use a web service
7 async function isPositive(text: string): Promise<boolean> {
8   const response = await fetch(`http://text-processing.com/api/sentiment/`, {
9     method: "POST",
10    body: `text=${text}`,
11    headers: {
12      "Content-Type": "application/x-www-form-urlencoded",
13    },
14  });
15  const json = await response.json();
16  return json.label === "pos";
17 }
```

At the bottom left of the editor, there is a blue button with the Copilot logo and the text 'Copilot'.

<https://github.com/features/copilot>



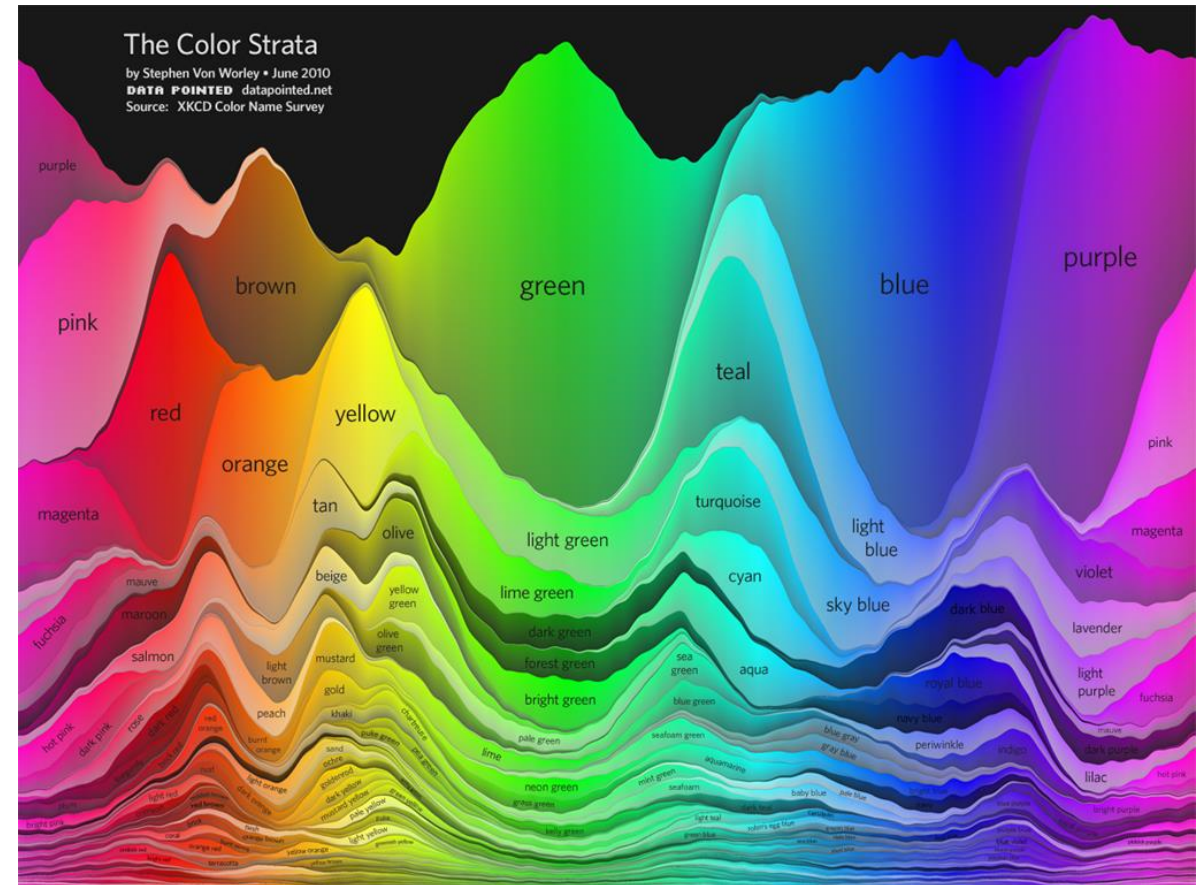
# What is language?

- Language is used to communicate
  - Things, actions, abstract concepts



# What is language?

- Language puts categories on the world
  - It discretizes a continuous space

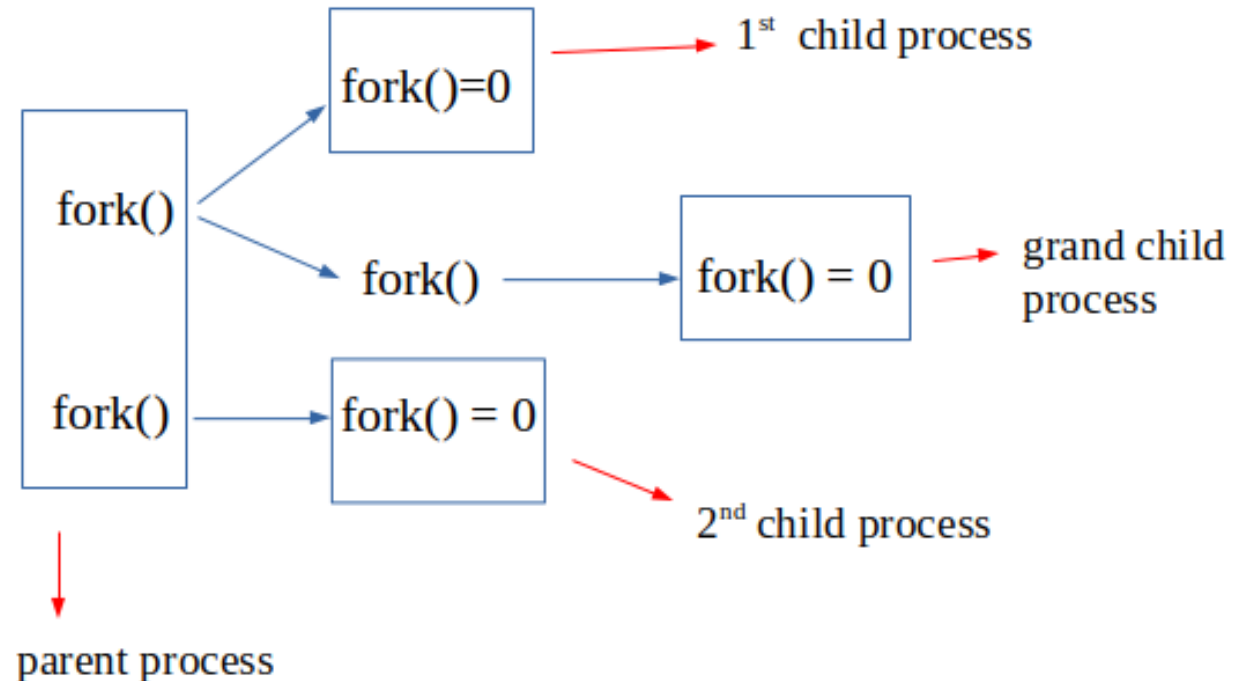




# What is language?

- Language picks out what is salient and important
- What concepts do we have words for?

## fork



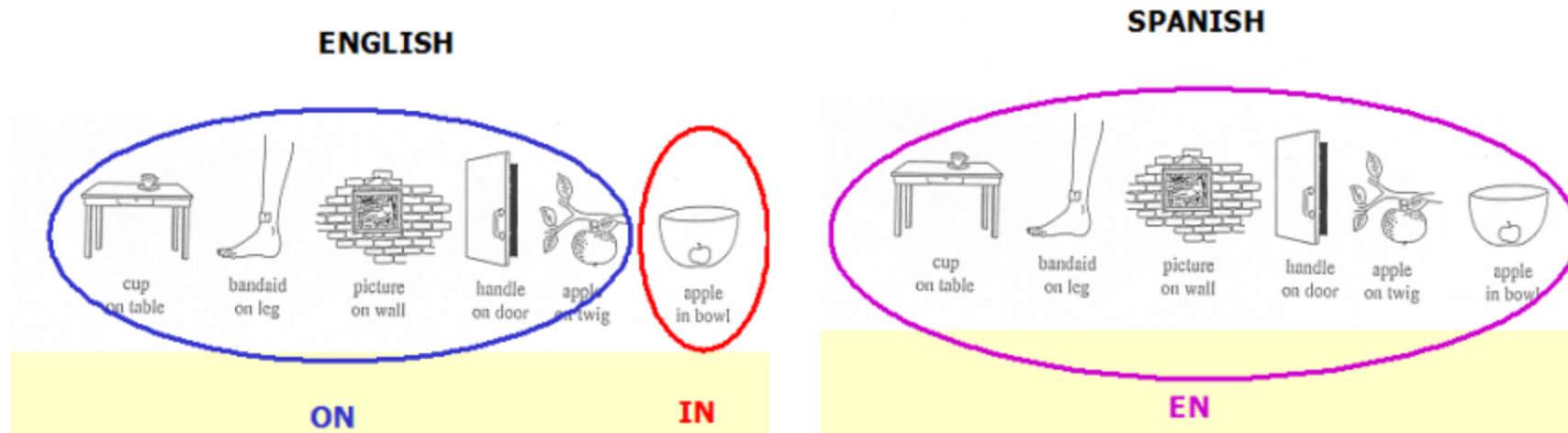
# What is language?

- Language picks out what is salient and important
- What concepts do we have words for?
- Different languages have different discretization boundaries



# What is language?

- Language picks out what is salient and important
- What concepts do we have words for?
- Different languages have different discretization boundaries



<http://pyersqr.org/classes/Ling731/Space2.htm>

# Natural Language Processing



Building **useful system** to process language

# Computational Linguistics



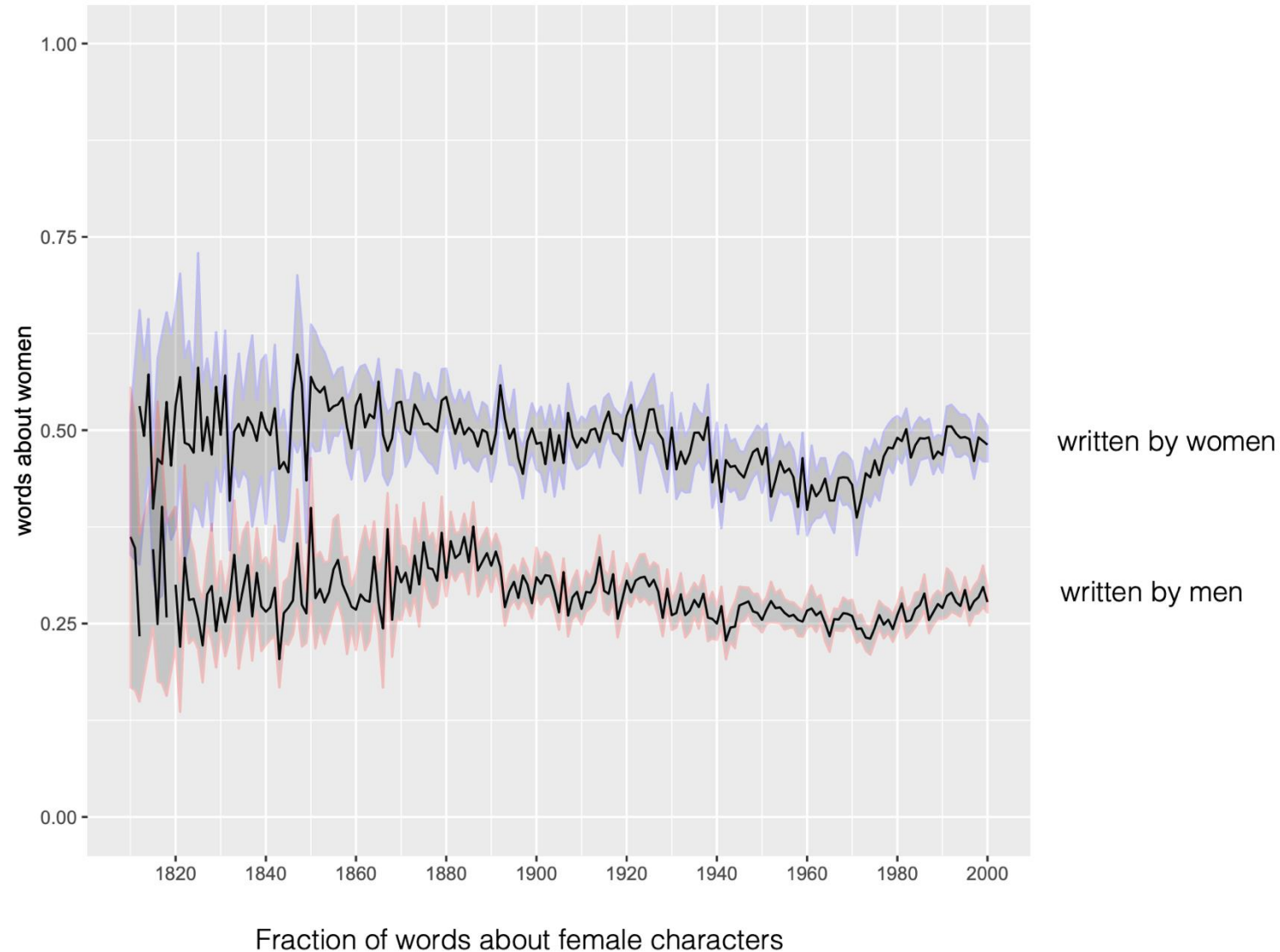
"I just had a heated conversation  
with my computer."

(image credit: <https://www.enterrasolutions.com/blog/computational-linguistics-and-natural-language-processing/>)

Using computers to study **human** language



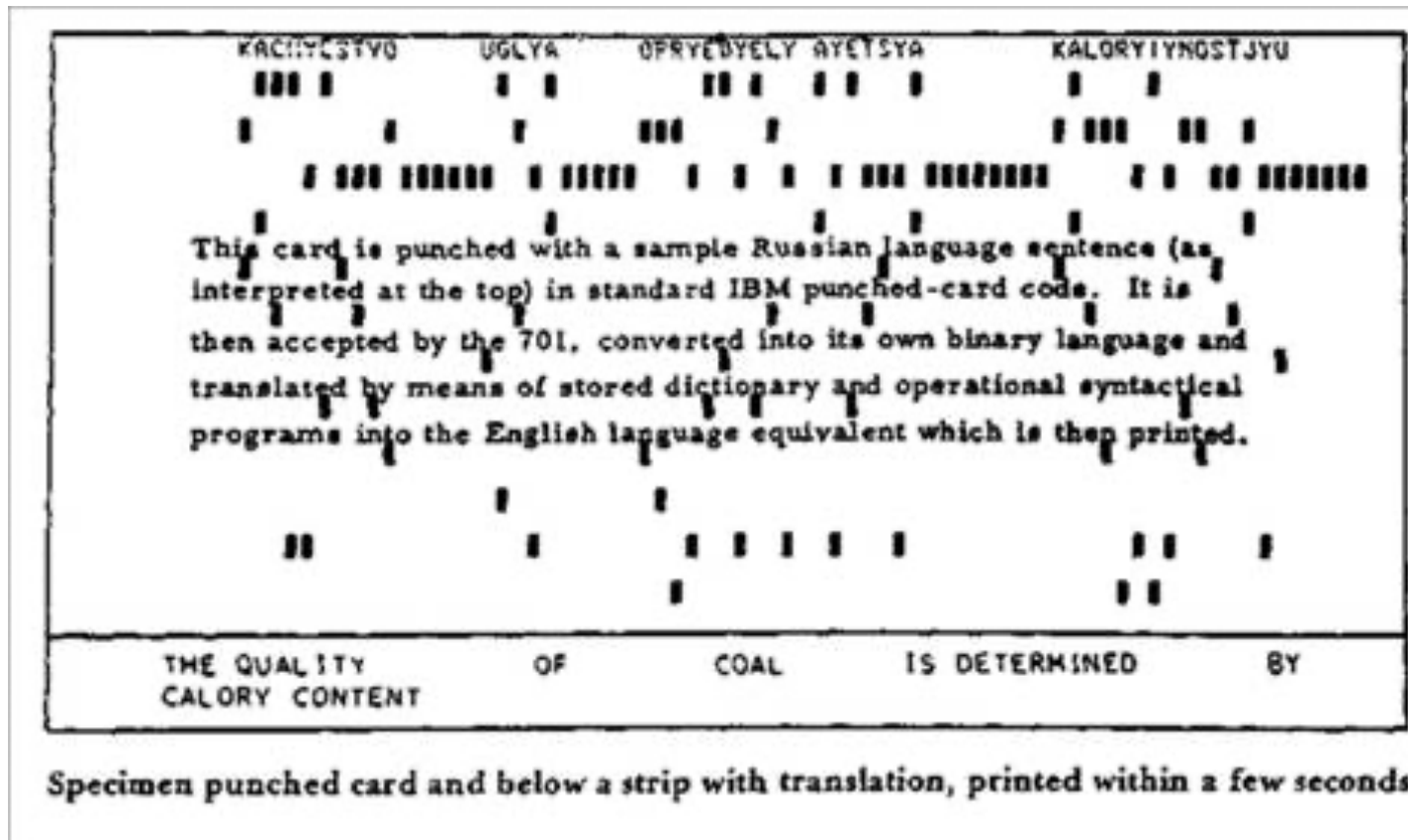
# Analyzing word usage in literature



Ted Underwood, David Bamman, and Sabrina Lee (2018),  
"The Transformation of Gender in English-Language Fiction," Cultural Analytics

# Brief history of NLP

# Beginnings



Georgetown-  
IBM  
experiment,  
1954

“Within three or five years, machine translation will be a solved problem”

# SHRDLU (Winograd, 1968)

Video of actual system: <https://www.youtube.com/watch?v=bo4RvYJYOzI>

Person: Pick up a big red block.

Computer: OK.

Person: Grasp the pyramid.

Computer: I don't understand which pyramid you mean.

Person (changing their mind): Find a block which is taller than the one you are holding and put it into the box.

Computer: By "it", I assume you mean the block which is taller than the one I am holding.

Computer: OK.

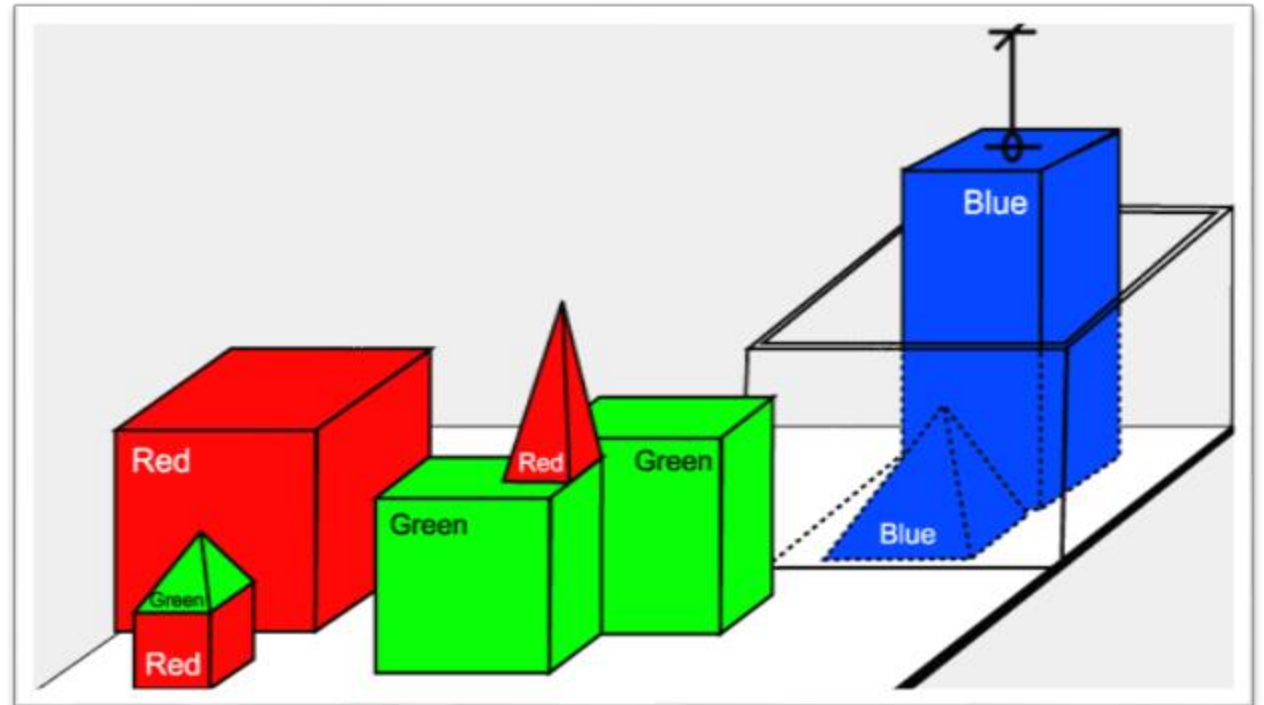
Person: What does the box contain?

Computer: The blue pyramid and the blue block.

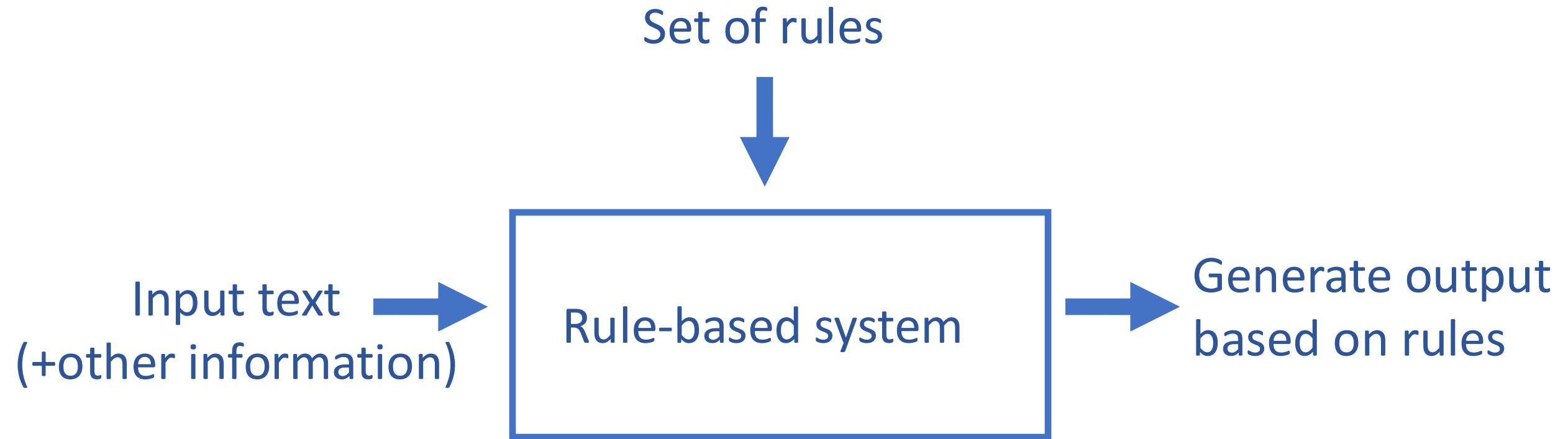
Person: What is the pyramid supported by?

Computer: The box.

Lots of rules!



# Using rules for NLP



Match a pattern

- Regular expressions

  - . \*ed => past tense verb

- Contain keyword: delicious

# Eliza (Weizenbaum, 1966)

Demo: <http://psych.fullerton.edu/mbirnbaum/psych101/Eliza.htm>

Men are all alike.

IN WHAT WAY

They're always bugging us about something or other.

CAN YOU THINK OF A SPECIFIC EXAMPLE

Well, my boyfriend made me come here.

YOUR BOYFRIEND MADE YOU COME HERE

He says I'm depressed much of the time.

I AM SORRY TO HEAR YOU ARE DEPRESSED

Rogarian psychologist:  
reflect back what the patient said

- Set of ranked transformation **rules** based on **keywords**
- Apply rules based on matching words in sentence



# Transformation rules

Keywords are ranked from specific to general

I know everybody laughed at me

- “I” is a very general keyword:

I: (I \*) -> (You say you 2)

YOU SAY YOU KNOW EVERYBODY LAUGHED AT YOU

- “Everybody” is much more interesting (someone using universals like everybody/always is probably “referring to some quite specific event or person”)

WHO IN PARTICULAR ARE YOU THINKING OF?

- Implementation: keywords stored with their rank

**Everybody** 5 (*transformation rules*)

**I** 0 (*transformation rules*)

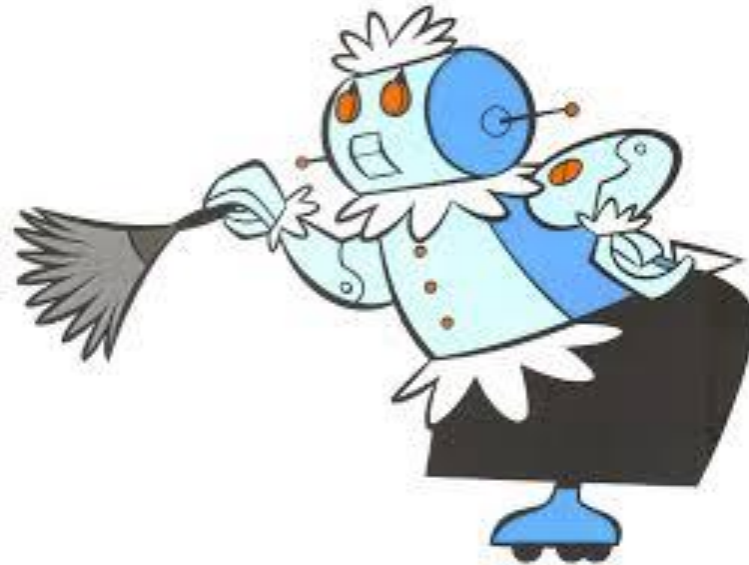
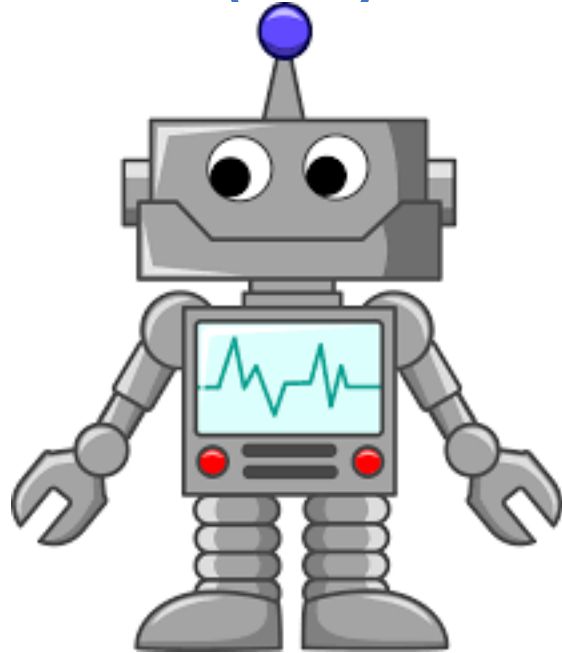
Backoff

Please go on

That's very interesting

I see

Where is my block stacking or  
housekeeper robot that I can talk to?  
(maybe we are getting close...)



Rosie from the Jetsons



Understanding  
language is hard!

The Far Side - Gary Larson

Why is natural language processing hard?

# Programming Languages

C, C++, Java, Python, ...

- Unambiguous
- Fixed
- Designed
- Learnable?
- Known simple semantics

# Natural Languages

French, English, Korean, Chinese, Tagalog, ...

- Ambiguous
- Evolving
- Transmitted
- Learnable
- Complex semantics

# Why is NLP hard?

Interpretation of language assumes a common basis of **world knowledge** and **context**

- **Ambiguous:**
  - “bank”, “bat”
  - “Milk Drinkers Turn to Powder”
- **Variation:** Many ways to say same thing
- **Context dependent:**
  - natural language is under-specified



Herb Clark

“bat’k”

**Table**



**Counter**

# Context-dependence

“I put the bowl on the **table**”



“The numbers in the **table** don’t add up”

Table 8.2 Calculation of the  $\chi^2$  test on figures in table 8.1

| Class<br>(1) | Expected numbers |          | O - E    |          | (O-E) <sup>2</sup> /E |          |
|--------------|------------------|----------|----------|----------|-----------------------|----------|
|              | A<br>(2)         | B<br>(3) | A<br>(4) | B<br>(5) | A<br>(6)              | B<br>(7) |
| I            | 11.80            | 10.20    | 5.20     | -5.20    | 2.292                 | 2.651    |
| II           | 24.67            | 21.33    | 0.33     | -0.33    | 0.004                 | 0.005    |
| III          | 39.15            | 33.85    | -0.15    | 0.15     | 0.001                 | 0.001    |
| IV           | 48.81            | 42.19    | -6.81    | 6.81     | 0.950                 | 1.009    |
| V            | 30.57            | 26.43    | 1.43     | -1.43    | 0.067                 | 0.077    |
| Total        | 30.57            | 134.00   | 0        | 0        | 3.314                 | 3.833    |

$$\chi^2 = 3.314 + 3.833 = 7.147, \text{ d.f.} = 4, 0.10 < P < 0.50.$$

“I made her duck”

I cooked her duck (the waterfowl) for dinner

I cooked the duck (the waterfowl) belonging to her.

I created a model of a duck that she now has.

I caused her to lower her head



# Some language humor (from real newspaper headlines)

Kids make nutritious snacks

Stolen painting found by tree

Miners refuse to work after death

Squad helps dog bite victim

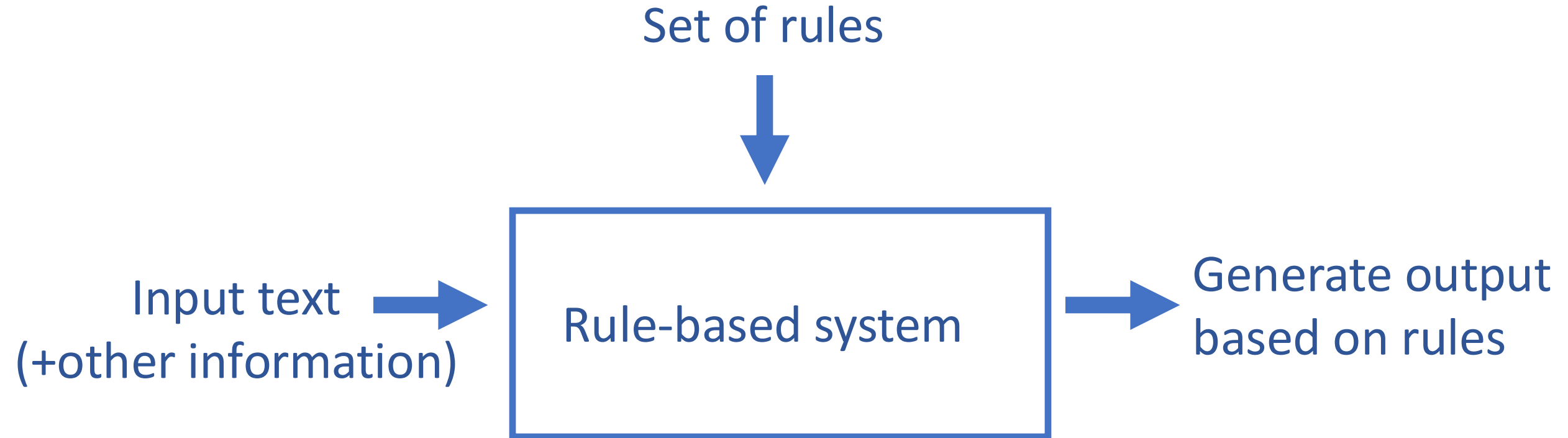
Killer sentenced to die for second time in 10 years

Lack of brains hinders research

Pooches and people digging newly opened park

# Using rules for NLP

- Translation with rules started in the 1950s
- Eliza chatbot (1966)
- SHRDLU grounded language understanding (1968)



# Rule-based Machine Translation

- Georgetown IBM translator
- Bilingual dictionary to map Russian words to their English counterparts
- But what if the word order is different? How to handle all the nuances of meaning?



1 minute video showing 1954 MT:

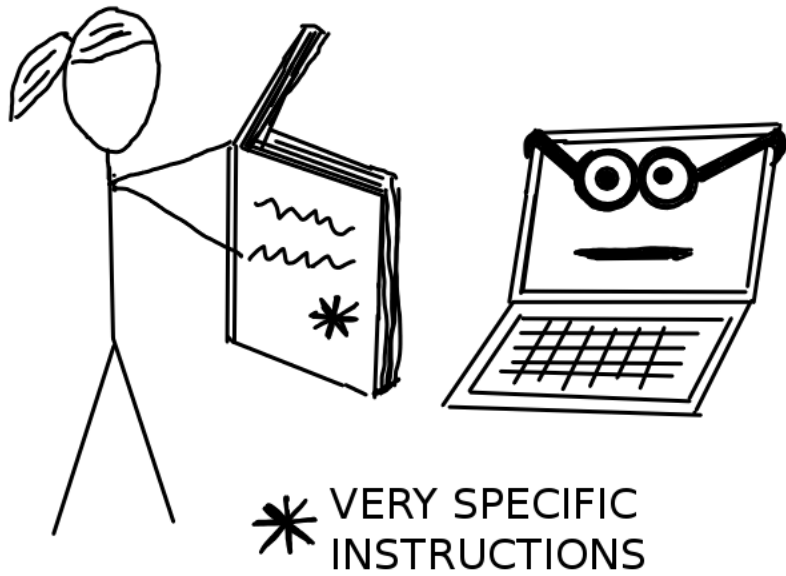
<https://youtu.be/K-HfpsHPmvw>

[https://www.ibm.com/ibm/history/exhibits/701/701\\_translator.html](https://www.ibm.com/ibm/history/exhibits/701/701_translator.html)

Coming up rules is hard!

Let's learn from data!

## Without Machine Learning



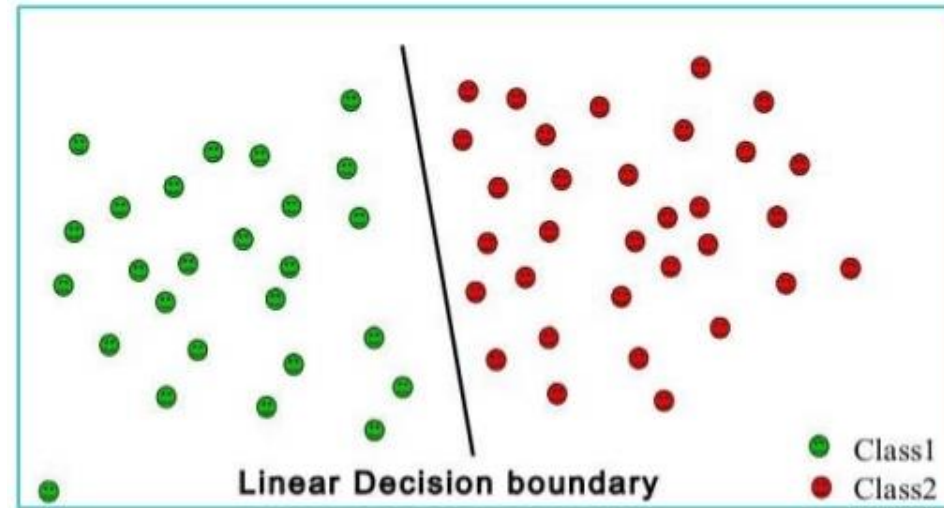
## With Machine Learning



<https://christophm.github.io/interpretable-ml-book/terminology.html>

# Rise of statistical learning

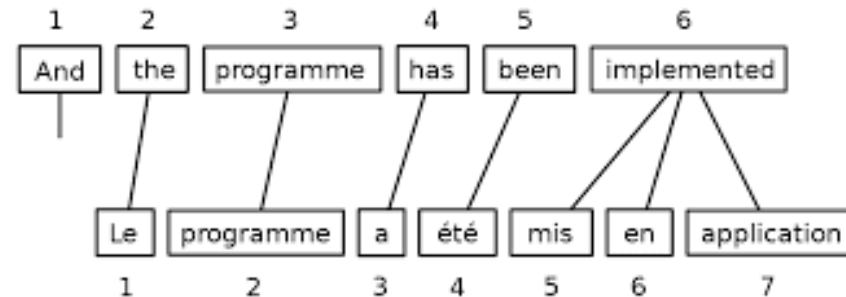
(late) 1980s to 2000s: Statistical MT, 2000s-2014: Statistical Phrase-Based MT



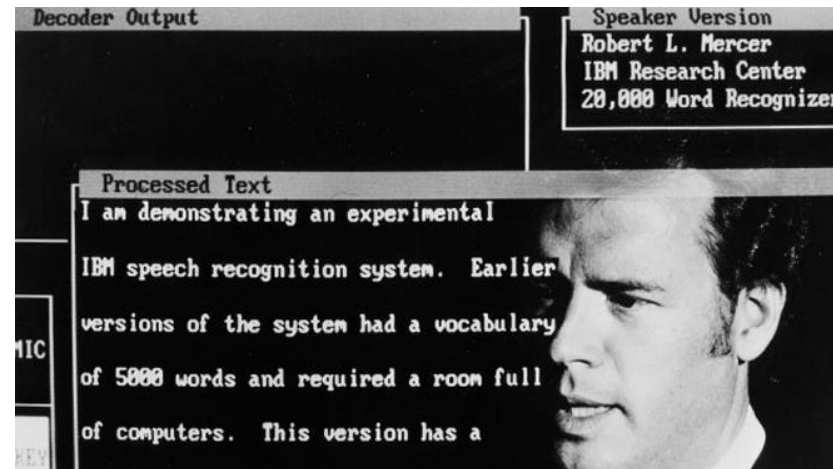
- Use of machine learning techniques in NLP
- Increase in computational capabilities
- Availability of electronic corpora
- Data-driven evaluation of models

# Rise of statistical learning

(late) 1980s to 2000s: Statistical MT, 2000s-2014: Statistical Phrase-Based MT



IBM Models  
for translation



Speech  
recognition

*Anytime a linguist leaves the group the (speech) recognition rate goes up*

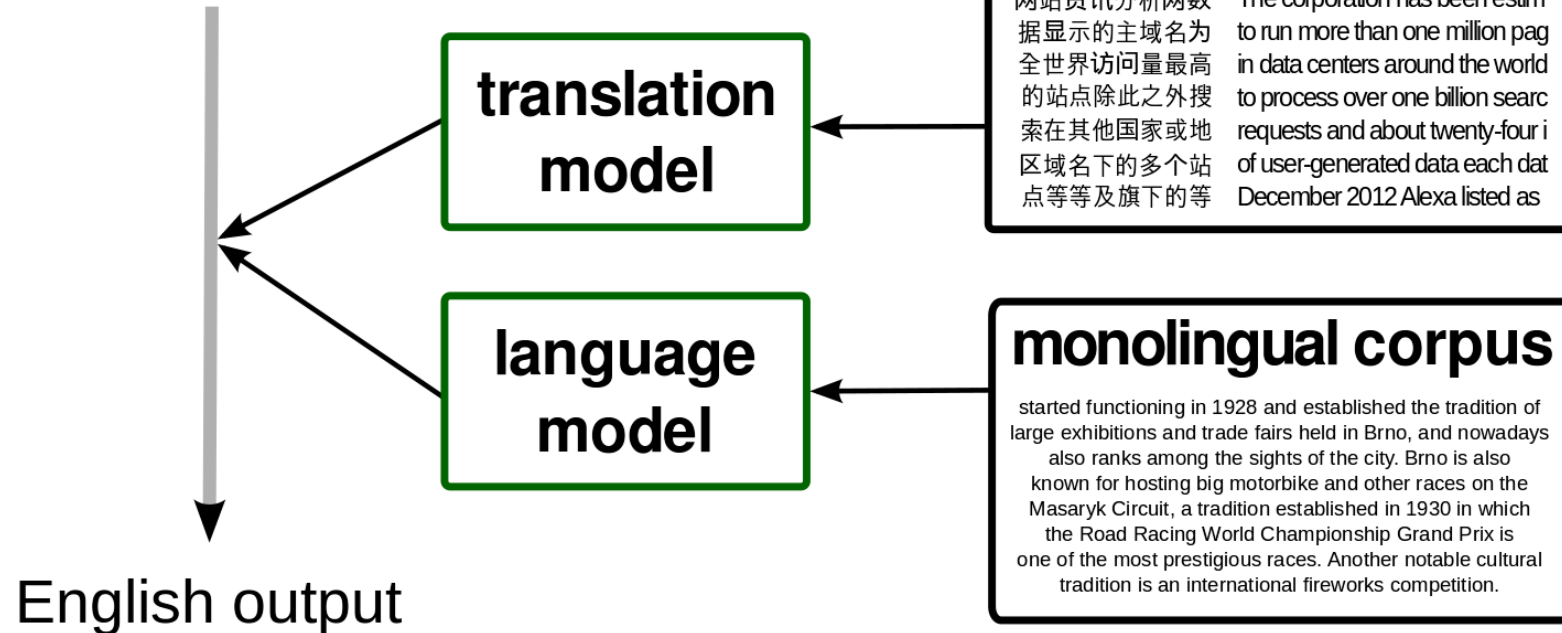
- Fred Jelinek



# Statistical Machine Translation

- Required a parallel corpora

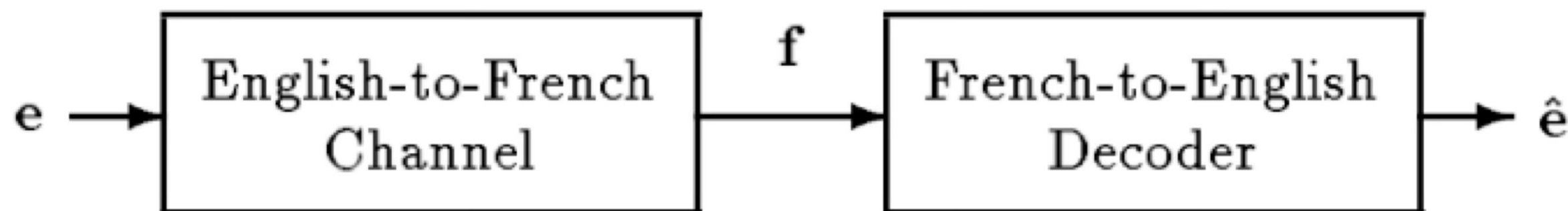
似乎格式有問題



# The Candide System for Machine Translation

*Adam L. Berger, Peter F. Brown\*, Stephen A. Della Pietra, Vincent J. Della Pietra,  
John R. Gillett, John D. Lafferty, Robert L. Mercer\*, Harry Printz, Luboš Ureš*

IBM Thomas J. Watson Research Center  
P.O. Box 704  
Yorktown Heights, NY 10598



$$\hat{e} = \operatorname{argmax}_e \Pr(e \mid f) = \operatorname{argmax}_e \Pr(f \mid e) \Pr(e)$$

Bayes Rule

# Rise of statistical learning

- Required human-designed representations and input features

| Var   | Definition                                                                            | Value in Fig. <a href="#">5.2</a> |
|-------|---------------------------------------------------------------------------------------|-----------------------------------|
| $x_1$ | count(positive lexicon) $\in$ doc)                                                    | 3                                 |
| $x_2$ | count(negative lexicon) $\in$ doc)                                                    | 2                                 |
| $x_3$ | $\begin{cases} 1 & \text{if "no"} \in \text{doc} \\ 0 & \text{otherwise} \end{cases}$ | 1                                 |
| $x_4$ | count(1st and 2nd pronouns $\in$ doc)                                                 | 3                                 |
| $x_5$ | $\begin{cases} 1 & \text{if "!"} \in \text{doc} \\ 0 & \text{otherwise} \end{cases}$  | 0                                 |
| $x_6$ | log(word count of doc)                                                                | $\ln(64) = 4.15$                  |

Can these features be automatically learned?

# Promise of deep learning

- Most NLP works in the past focused on human-designed representations and input features

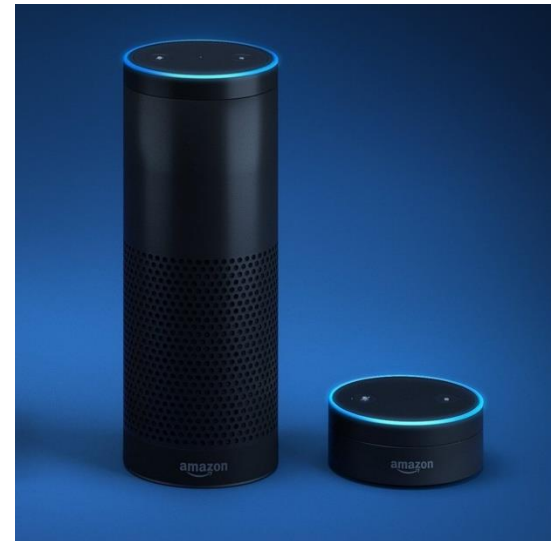
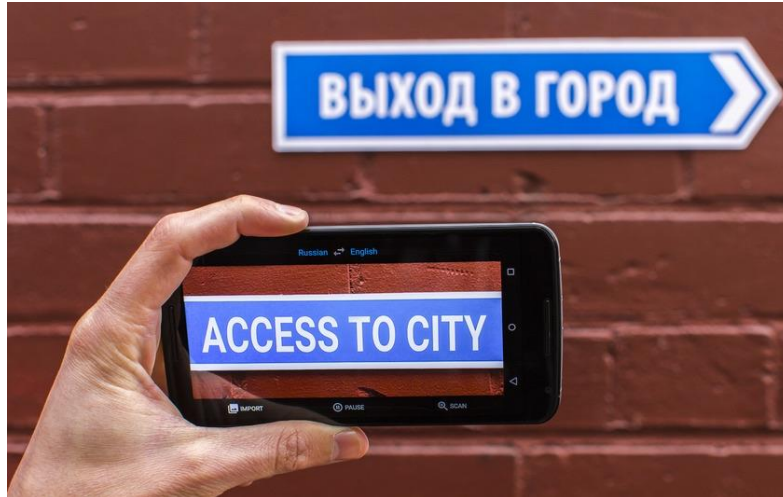
| Var   | Definition                                                                            | Value in Fig. 5.2 |
|-------|---------------------------------------------------------------------------------------|-------------------|
| $x_1$ | count(positive lexicon) $\in$ doc)                                                    | 3                 |
| $x_2$ | count(negative lexicon) $\in$ doc)                                                    | 2                 |
| $x_3$ | $\begin{cases} 1 & \text{if "no"} \in \text{doc} \\ 0 & \text{otherwise} \end{cases}$ | 1                 |
| $x_4$ | count(1st and 2nd pronouns $\in$ doc)                                                 | 3                 |
| $x_5$ | $\begin{cases} 1 & \text{if "!"} \in \text{doc} \\ 0 & \text{otherwise} \end{cases}$  | 0                 |
| $x_6$ | log(word count of doc)                                                                | $\ln(64) = 4.15$  |

- **Representation learning attempts to automatically learn good features and representations**
- **Deep learning attempts to learn multiple levels of representation on increasing complexity/abstraction**

# Deep learning era

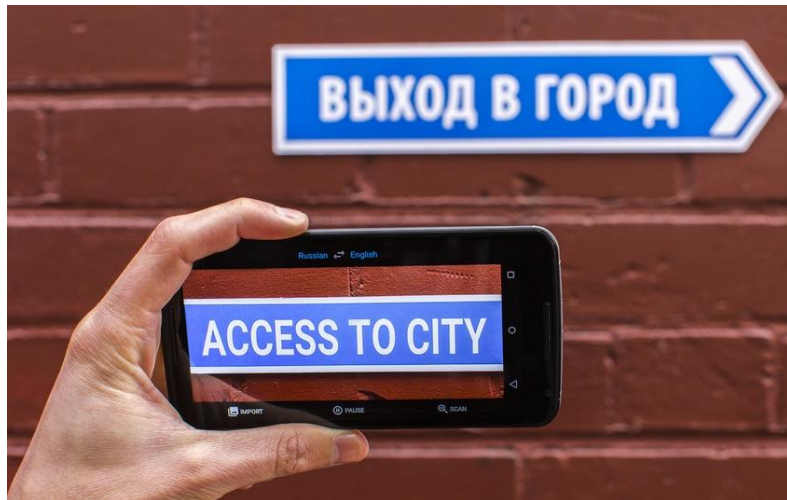
2014-Present: Neural Machine Translation

- Led to significant advances in core NLP technologies



# Deep learning era

- Significant advances in core NLP technologies
- **Essential ingredient:** large-scale supervision, lots of compute
- Reduced manual effort - less/zero feature engineering



36M sentence pairs

*Russian: Машинный перевод - это круто!*



*English: Machine translation is cool!*



# Machine Translation

BBC

Sign in

选项 (英文)

检索

NEWS | 中文

繁

主页 | 国际 | 两岸 | 英国 | 评论 | 科技 | 财经 | 图辑 | 音频材料 | 视频材料 | BBC英伦网

巴拿马首任驻华大使专访：与台湾断交之后

他说，不认同中国“买走”台湾邦交国的说法，与中国建立关系对巴拿马有利，不担心影响与美国关系。

🕒 1小时前

巴拿马外交转向周年 中美大国博弈内幕解密

尼加拉瓜运河成谜：人走楼空 “不再提及”

触发萨尔瓦多与台湾断交的港口令美国担忧



特别推荐



人民币贬值、楼价高涨：消费降级中产失去的优质生活

广告



观点：高铁“一地两检”——中国的强势 港人的无力

江浙：两地价值观和制度差异巨大，冲突无可避免。香港没有反对的权利，也没有反对的能力。



范冰冰消失百日后 中国娱乐业的寒蝉效应

日前发布的《中国影视明星社会责任研究报告》中，最高分徐峥为78分，最低分范冰冰为0分。

🕒 2小时前

中蒙参加俄罗斯军演 “中俄靠拢论”再吸眼球

🕒 2018年 9月 11日

日本人脚踹慰安妇铜像引爆台湾人抗议

🕒 2018年 9月 11日

太空望远镜探测比“三



台湾民间发起东京奥运“正名”公投的意义



# Machine Translation

**BBC** Sign in Option (English) 检索

**NEWS | 中文** Traditional

Homepage International Cross-strait United Kingdom comment Technology Finance Picture series Audio material

## Interview with Panama's first ambassador to China: After breaking diplomatic relations with Taiwan

He said that he does not agree with China's saying that "buy" Taiwan's diplomatic relations with China. Establishing relations with China is beneficial to Panama and does not worry about affecting relations with the United States.

🕒 1 hour ago

Panamanian Diplomacy Turns to Anniversary

The Nicaragua Canal is a mystery: people go to the floor and "no longer mention"

The port that triggered the break of El Salvador and Taiwan has worried the United States



#VISITPanamá



### Special recommendation

Renminbi depreciation, property prices are rising: consumption downgrades the loss of quality life in the middle class

ADVERTISING



### The significance of the Taiwanese people's "referred to" referendum



### China and Mongolia participate in the Russian military



# Google's Neural Machine Translation System: Bridging the Gap between Human and Machine Translation

<https://ai.googleblog.com/2016/09/a-neural-network-for-machine.html>

| <i><b>Input sentence:</b></i>             | <i><b>Translation (PBMT):</b></i>                                                                                                                                     | <i><b>Translation (GNMT):</b></i>                                                                                                                      | <i><b>Translation (human):</b></i>                                                                                                                                                |
|-------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 李克強此行將啟動中加總理年度對話機制，與加拿大總理杜魯多舉行兩國總理首次年度對話。 | Li Keqiang premier added this line to start the annual dialogue mechanism with the Canadian Prime Minister Trudeau two prime ministers held its first annual session. | Li Keqiang will start the annual dialogue mechanism with Prime Minister Trudeau of Canada and hold the first annual dialogue between the two premiers. | Li Keqiang will initiate the annual dialogue mechanism between premiers of China and Canada during this visit, and hold the first annual dialogue with Premier Trudeau of Canada. |

(Wu et al., 2016)

# Google's Neural Machine Translation System: Bridging the Gap between Human and Machine Translation

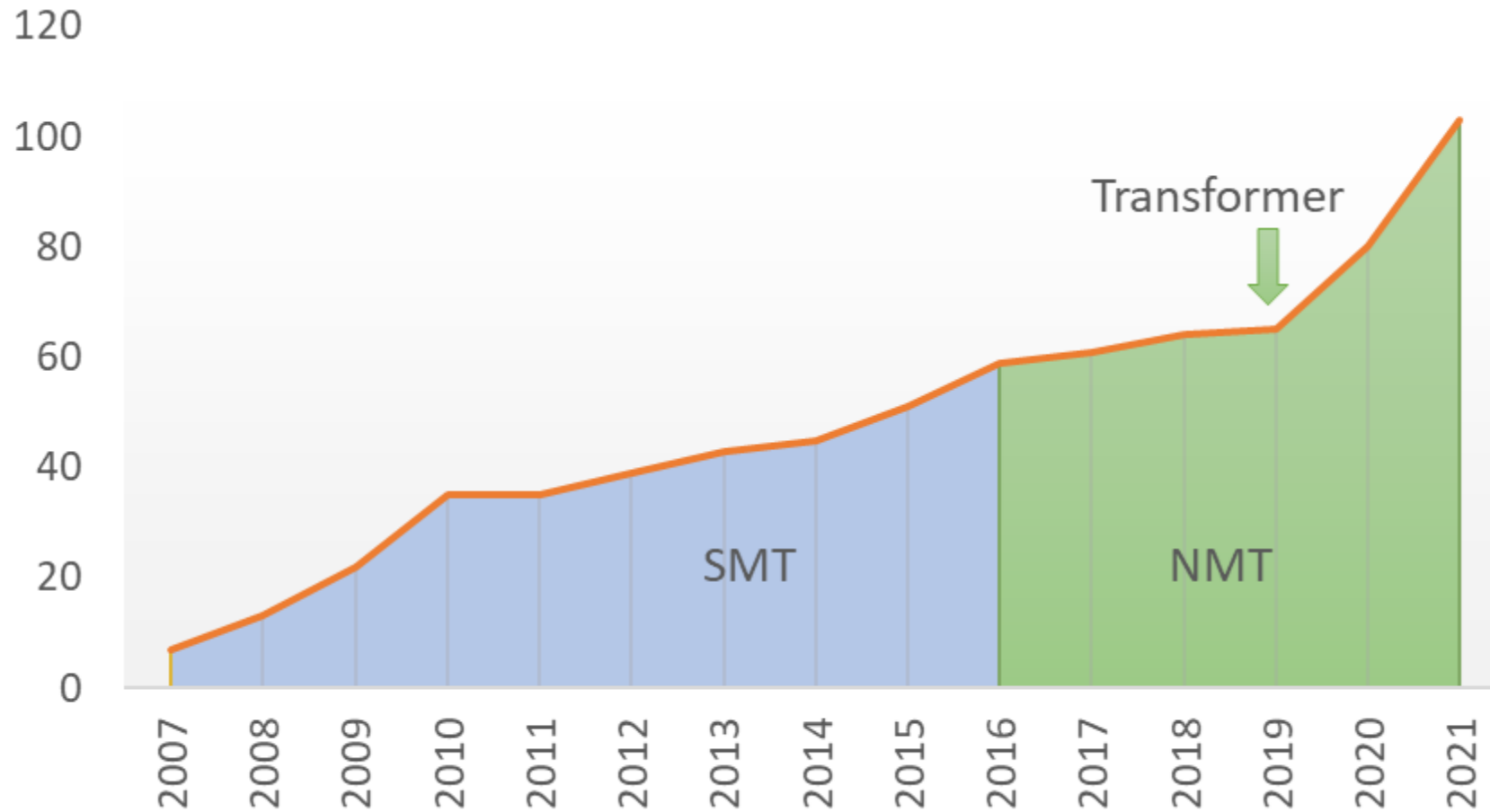
<https://ai.googleblog.com/2016/09/a-neural-network-for-machine.html>

Table 10: Mean of side-by-side scores on production data

|                   | PBMT  | GNMT  | Human | Relative Improvement |
|-------------------|-------|-------|-------|----------------------|
| English → Spanish | 4.885 | 5.428 | 5.504 | 87%                  |
| English → French  | 4.932 | 5.295 | 5.496 | 64%                  |
| English → Chinese | 4.035 | 4.594 | 4.987 | 58%                  |
| Spanish → English | 4.872 | 5.187 | 5.372 | 63%                  |
| French → English  | 5.046 | 5.343 | 5.404 | 83%                  |
| Chinese → English | 3.694 | 4.263 | 4.636 | 60%                  |

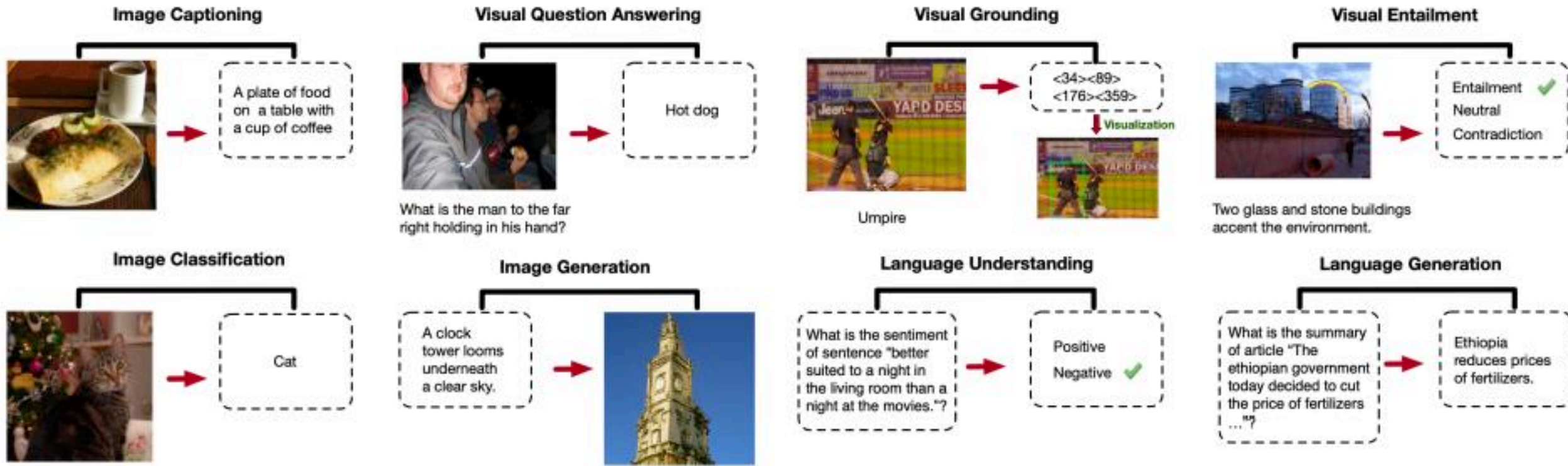
(Wu et al., 2016)

# Machine Translation (across 100 languages)



<https://www.microsoft.com/en-us/research/blog/microsoft-translator-now-translating-100-languages-and-counting>

# Multimodal models



<https://github.com/OFA-Sys/OFA>

OFA (Wang et al, 2022)

# Power of Data

## CleverBot (2010)

|            |                                      |
|------------|--------------------------------------|
| User:      | Do you like Doctor Who?              |
| Cleverbot: | Yes, so funny.                       |
| User:      | What's your favorite show?           |
| Cleverbot: | I don't have one.                    |
| User:      | What's your name?                    |
| Cleverbot: | Jason I already told you.            |
| User:      | No your real name.                   |
| Cleverbot: | Stephanie IS my real name.           |
| User:      | I thought it was Jason?              |
| Cleverbot: | How did you come to that conclusion. |
| User:      | Because you said so.                 |
| Cleverbot: | I have to do my homework.            |

<https://www.cleverbot.com/>

## How it works:

- Corpus of conversational turns
- Find the most similar sentence and copy the response
- Learn from human input

## What do you get?

- Something that someone say
- Incoherent conversation



# Power of Data

Meena (Google, 2020)

## Conversation D

---

**Human:** *I like how cute cows are*  
**Meena:** *Ah, interesting. You like how they are cute, right?*  
**Human:** *yeah, and they're really smart*  
**Meena:** *I heard they go to college*  
**Human:** *cows go to college?*  
**Meena:** *I heard that a cow went to Harvard.*  
**Human:** *what did the cow study?*  
**Meena:** *Bovine sciences.*  
**Human:** *do horses go to Harvard?*  
**Meena:** *Horses go to Hayvard.*  
**Human:** *that's a pretty good joke, I feel like you led me into it*  
**Meena:** *You were trying to steer it elsewhere, I can see it.*

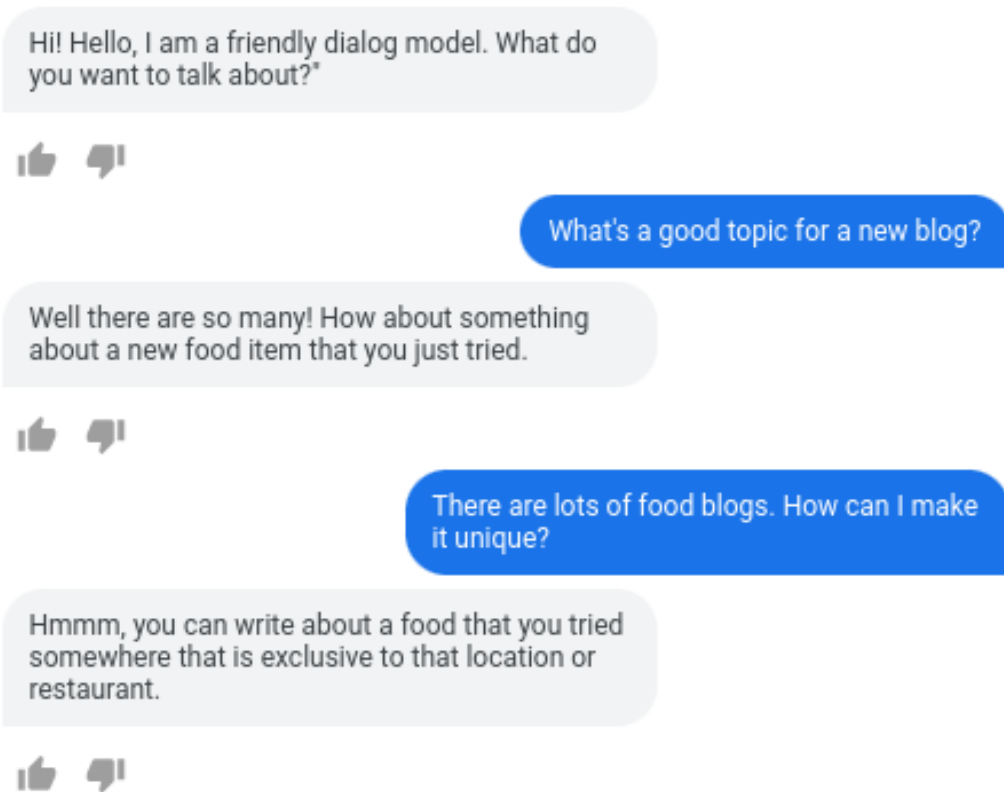
---

How it works:

- Corpus of conversational turns (over 40B words)
- Train huge neural network (2.6 billion parameters) for 30 days on 2048 TPUs cores
- Predict response given a sentence

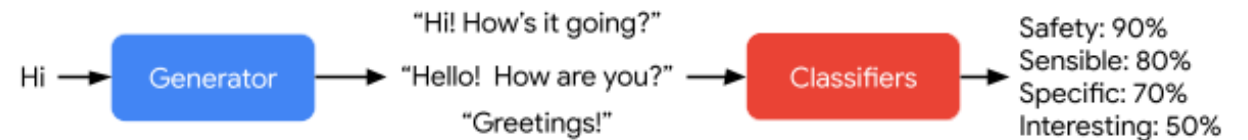
# Power of Data

LaMDA (Google, 2022)



How it works:

- Pretrain on even more data
- Fine-tune so it has certain characteristics



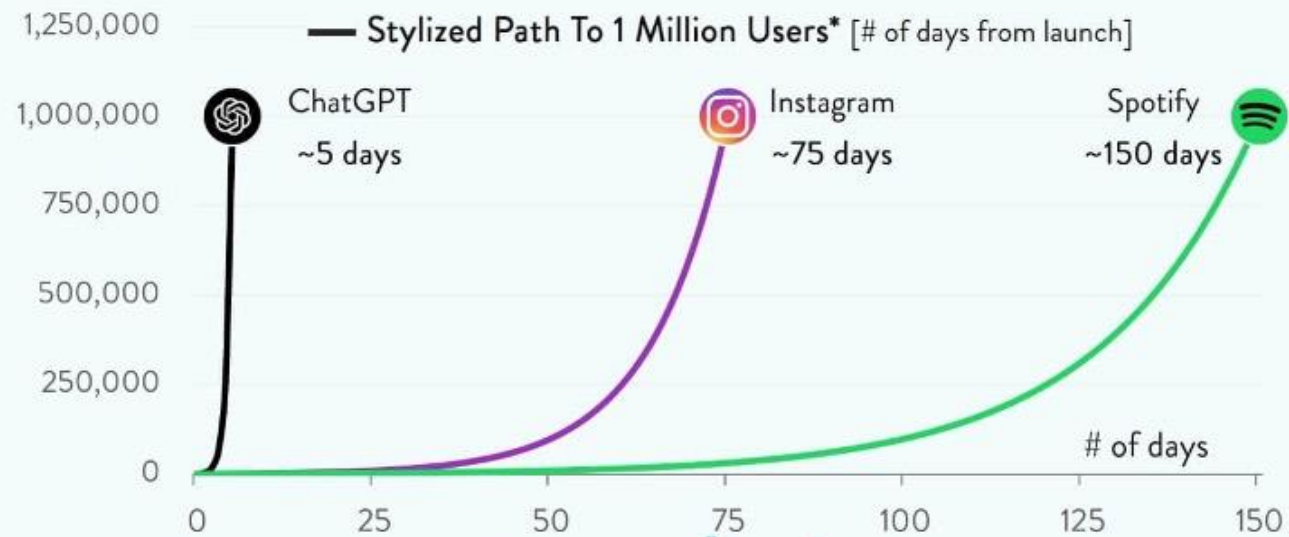
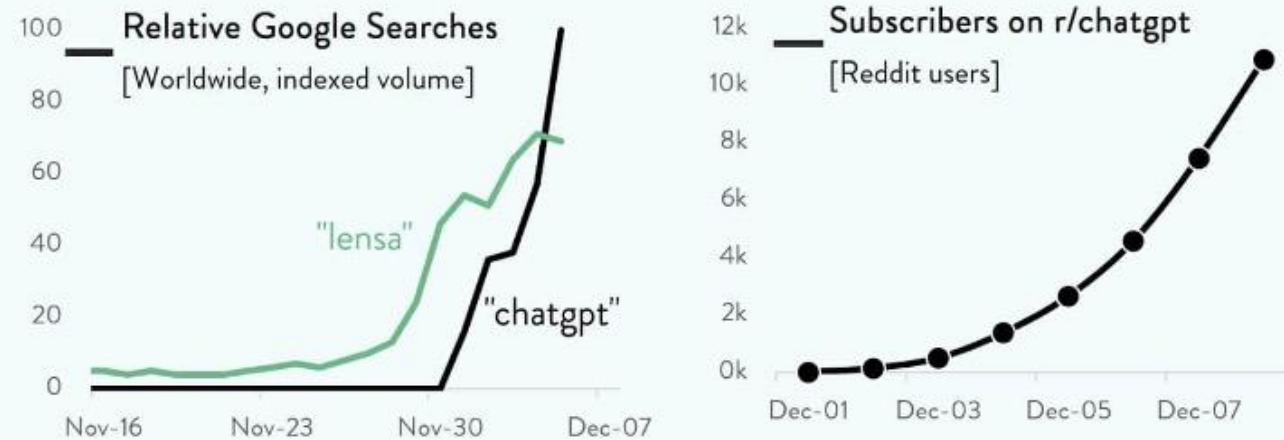
Is this chatbot sentient?

<https://ai.googleblog.com/2022/01/lamda-towards-safe-grounded-and-high.html>

[https://youtu.be/CvD9U\\_3hBPo](https://youtu.be/CvD9U_3hBPo)



# ChatGPT From OpenAI Is A Bot Taking The Tech World By Storm



Sources: Google, Subredditstats, Media Reports

chart\*Path is stylized to the 1m milestone

Source: Chart

# Power of Data

## ChatGPT (OpenAI, 2022)

<https://openai.com/blog/chatgpt/>



write a sonnet about pokemon



A world of wonder, filled with creatures wild,  
A realm of magic, where adventures start,  
From Kanto to Kalos, far and wide,  
The world of Pokémon, alive and heart.

With Pikachu as guide, we set out to roam,  
To catch and train them all, to be the best,  
Battling Gym Leaders, overcoming foes,  
Our bond with creatures strong, a bond we'll never test.

But it's not just battles that define,  
The world of Pokémon, so full of cheer,  
It's friendships formed, and lessons learned,  
A world that's full of love, and never fear.

So let's go on a journey, you and I,  
Through the world of Pokémon, beneath the sky.

## How it works:

- Trained on conversational data
- Uses reinforcement learning with human feedback

## *The Brilliance and Weirdness of ChatGPT*

A new chatbot from OpenAI is inspiring awe, fear, stunts and attempts to circumvent its guardrails.

<https://www.nytimes.com/2022/12/05/technology/chatgpt-ai-twitter.html>

Time taken to reach 1m users (mths)

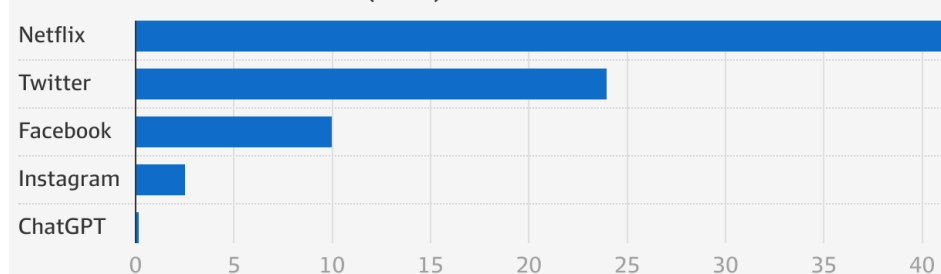
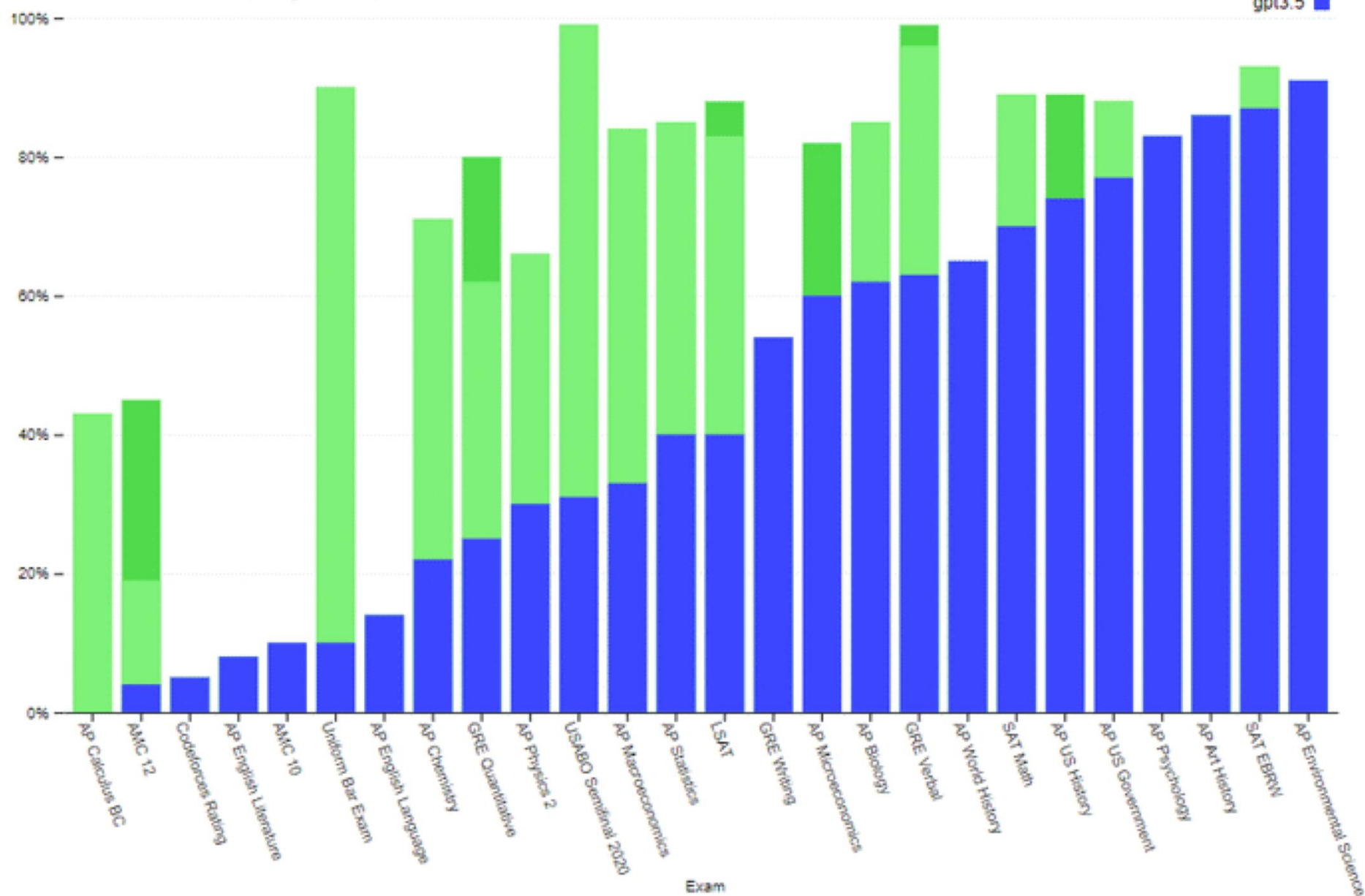


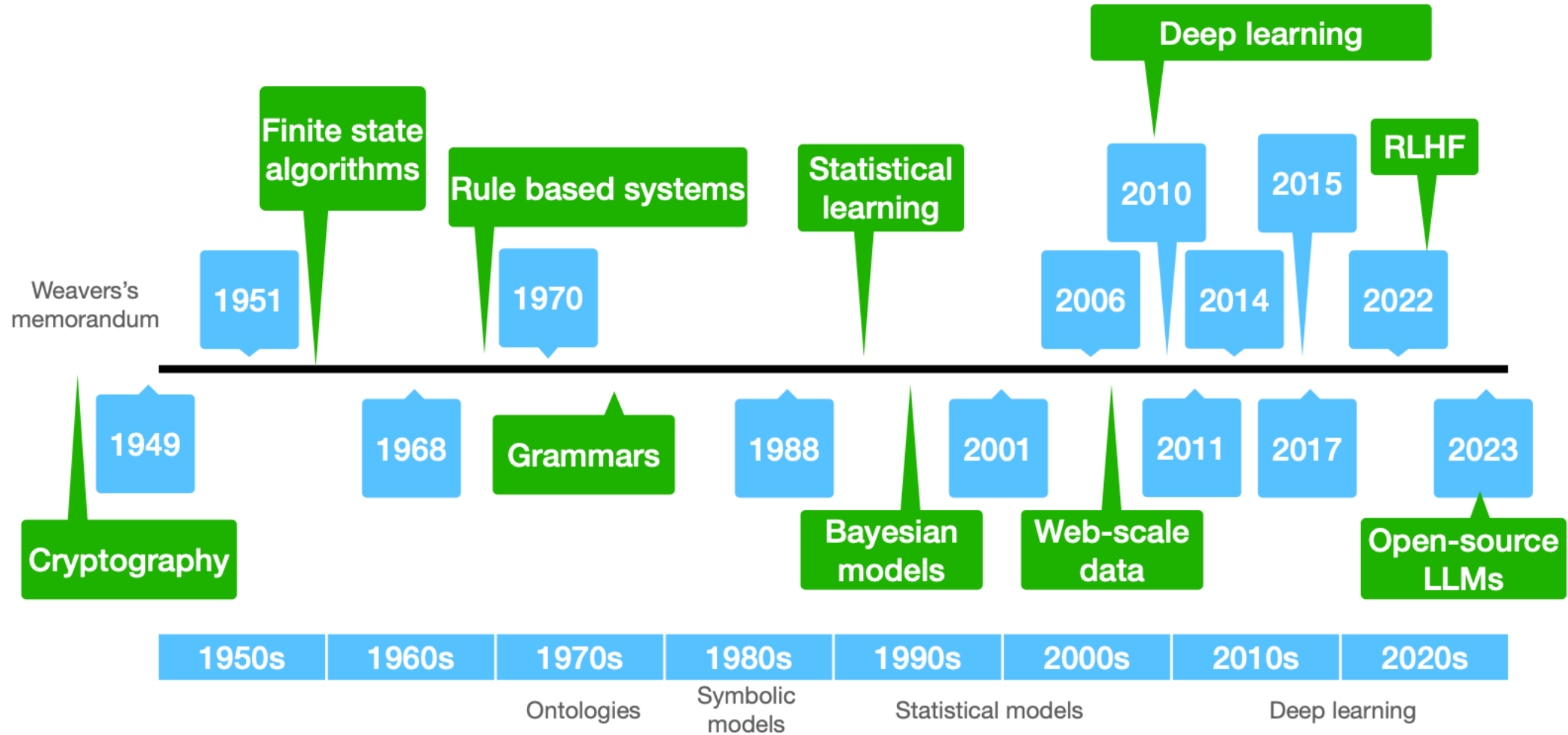
Chart: Financial Review • Source: Genevieve Roch-Decter, CFA

## Exam results (ordered by GPT-3.5 performance)

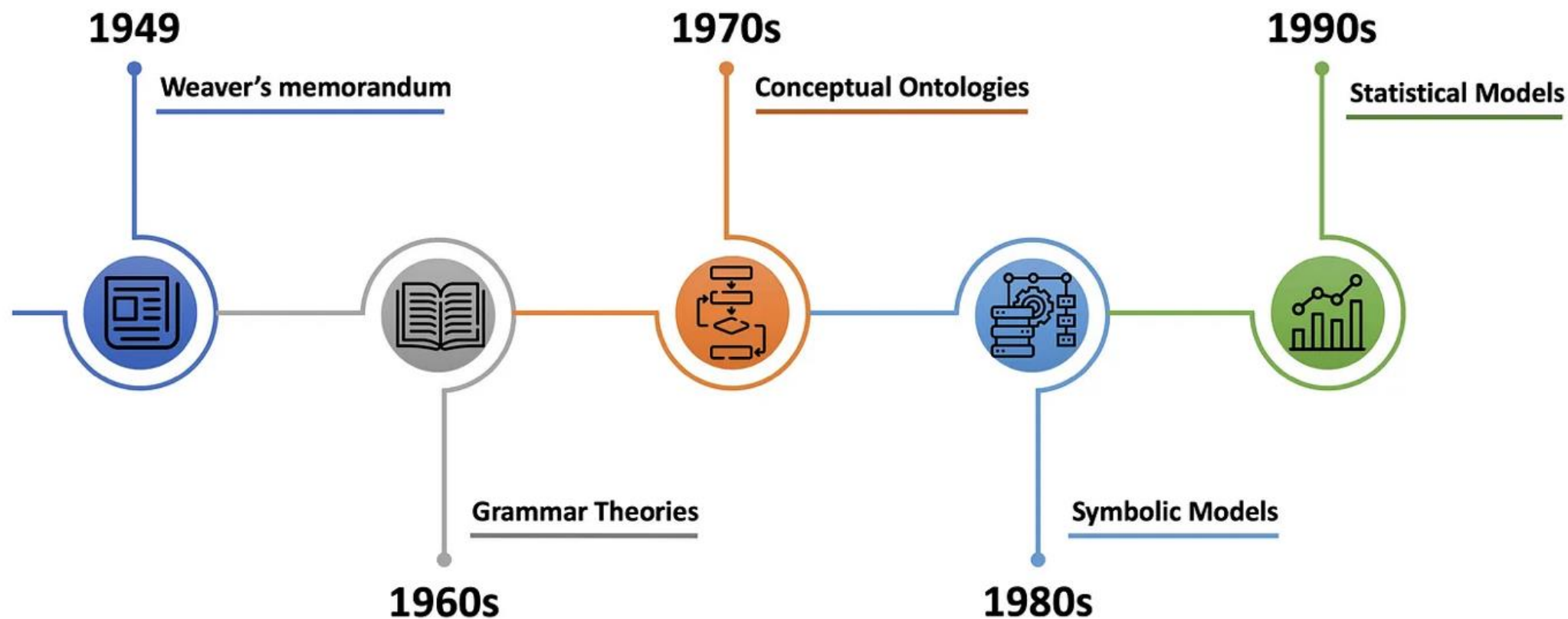
Estimated percentile lower bound (among test takers)



# Key developments in NLP

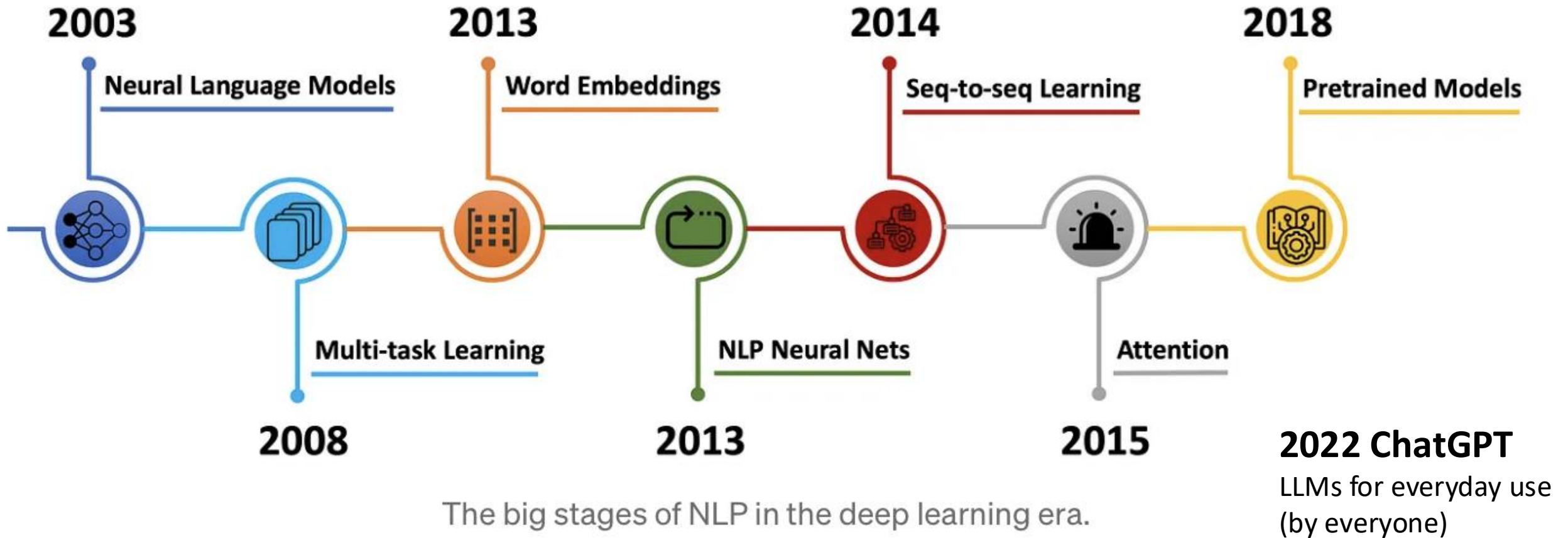


# NLP before deep learning



The big stages of NLP before the deep learning era.

# NLP in the deep learning era

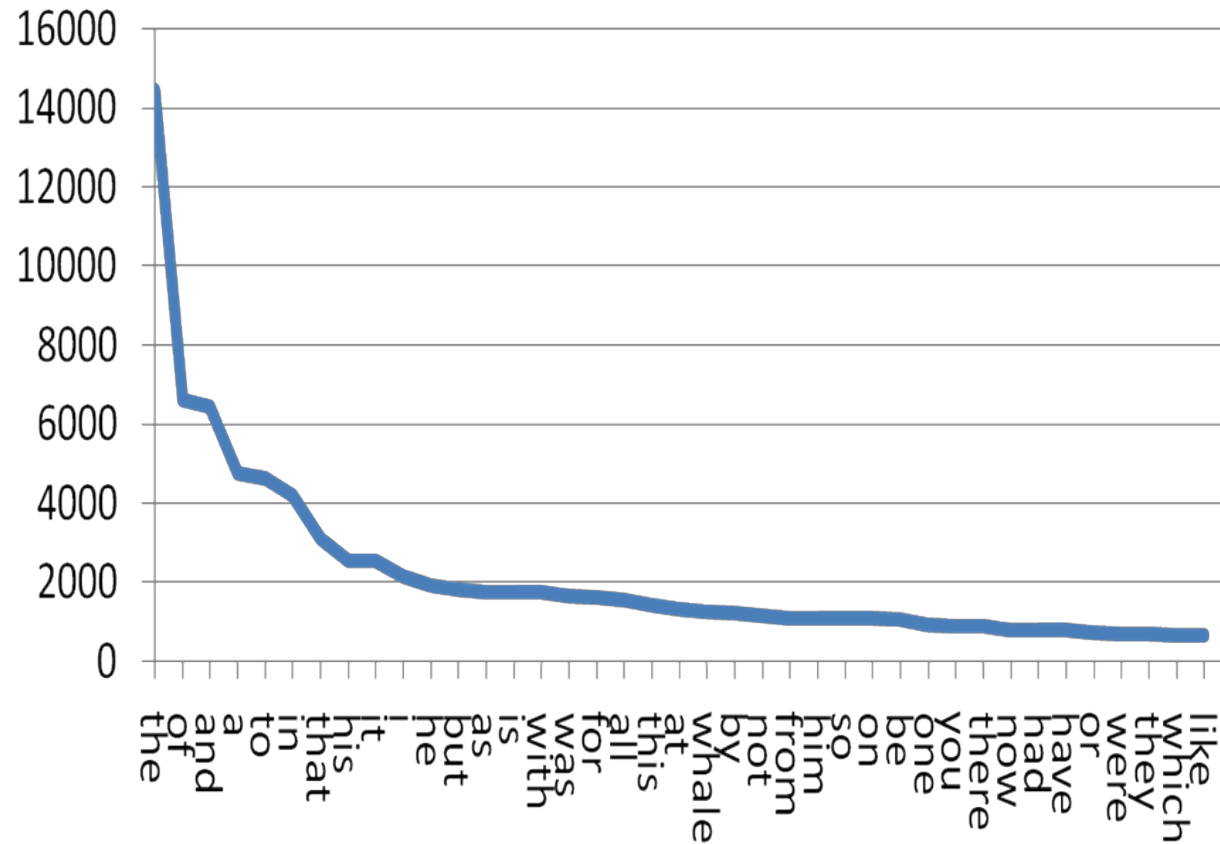


Many remaining challenges in NLP!



# Challenges in modern NLP

- *Scale*: Large number of phenomena
- *Sparsity*: Text data is often heavy-tailed



# Ambiguity in grounding

I saw her duck



# Challenges in modern NLP

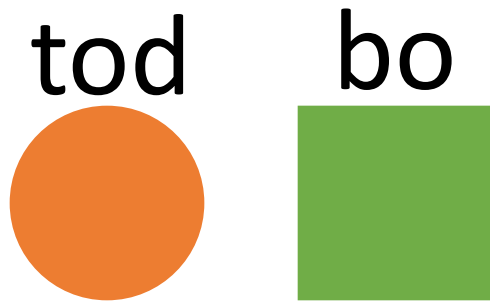
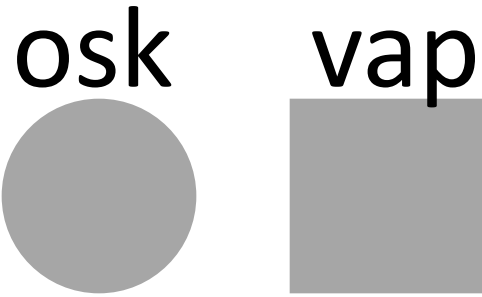
- *Bias*: Models learn biases in available data



- *Context*: Knowledge bases, perception, interaction

# Compositional generalization

Grounding



Compositionality



Generalization

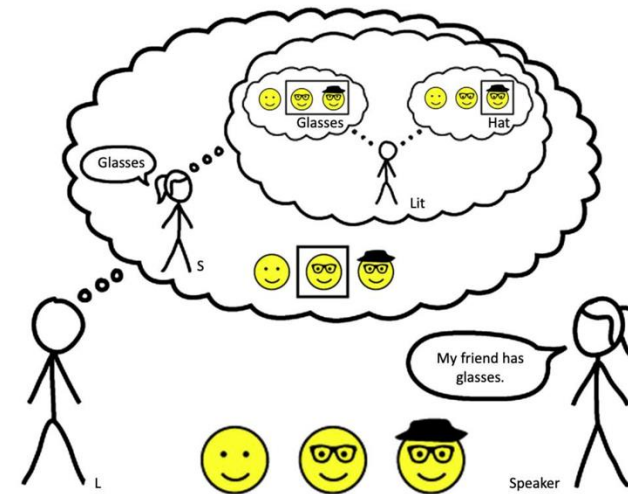


# Challenges of understanding language

- Does solving language require solving AI?
- Language derives its meaning from use



*'block' 'pillar' 'slab' 'beam'.*



- To truly understand language: need **context**, **world knowledge**, **mental model** of other people

# Grounded natural language

Move **OBJECT** from **START** to **GOAL**.

Move *the stuffed animal* from *the chair* to *the sofa*.



 Meta

HomeRobot: Open Vocabulary Mobile Manipulation

[Yenamandra et al. CoRL 2023]

<https://ovmm.github.io>

What will this course cover?



# What will this course cover?

## Representations

- Embeddings
- Compositional
- Structured

## Methods

- Rule based
- Statistical Methods
- Neural Models
- Dynamic programming
- Language Models

## Applications and Tasks

- Machine Translation
- Dialogue Agents
- Information Extraction
- Question Answering
- Grounding

# What will this course cover?

## Representations

- Embeddings
- Compositional
- Structured

## Methods

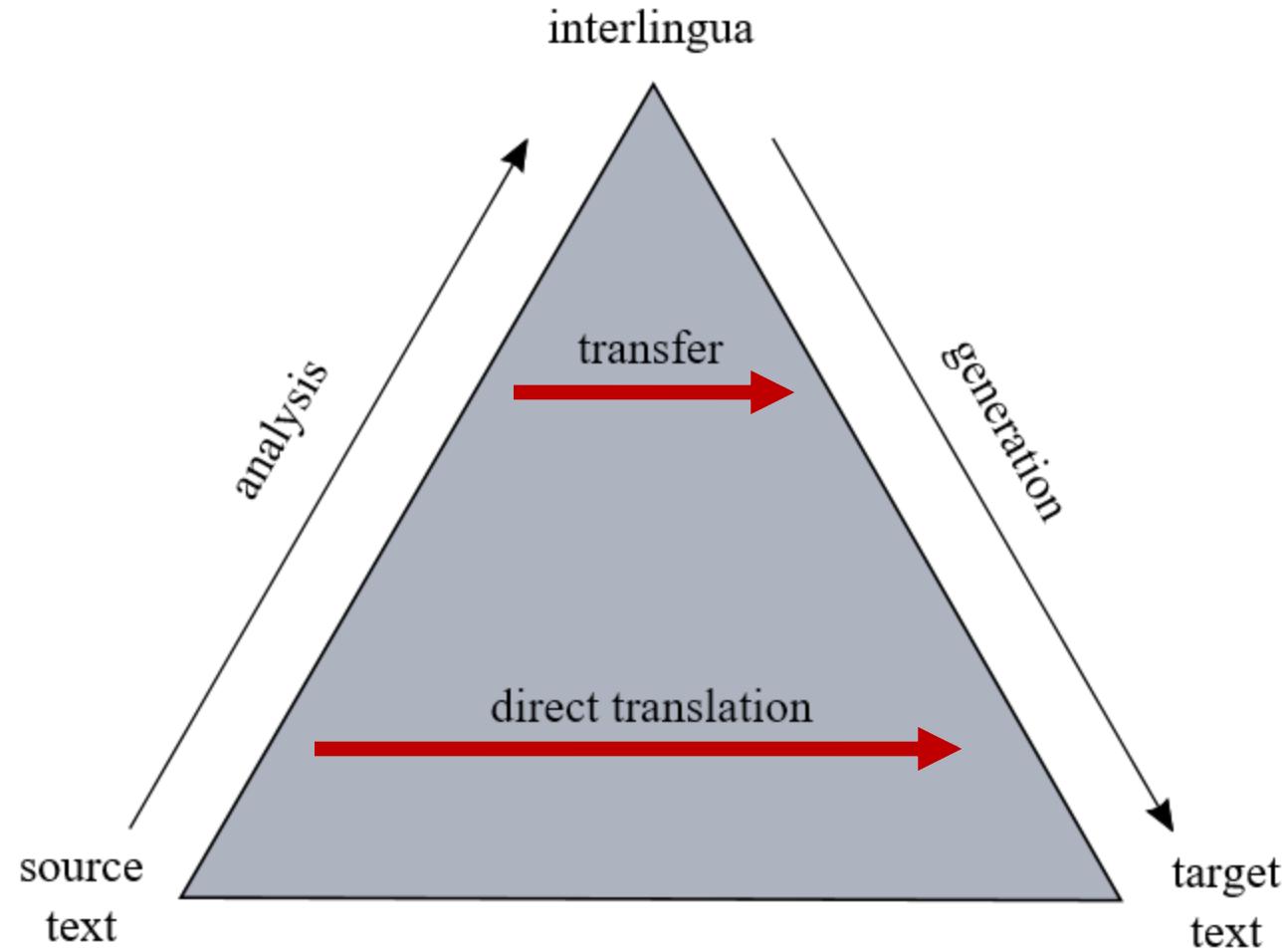
- Rule based
- Statistical Methods
- Neural Models
- Dynamic programming
- Language Models

## Applications and Tasks

- Machine Translation
- Dialogue Agents
- Information Extraction
- Question Answering
- Grounding

later part of the course

# Vauquois Triangle for translation



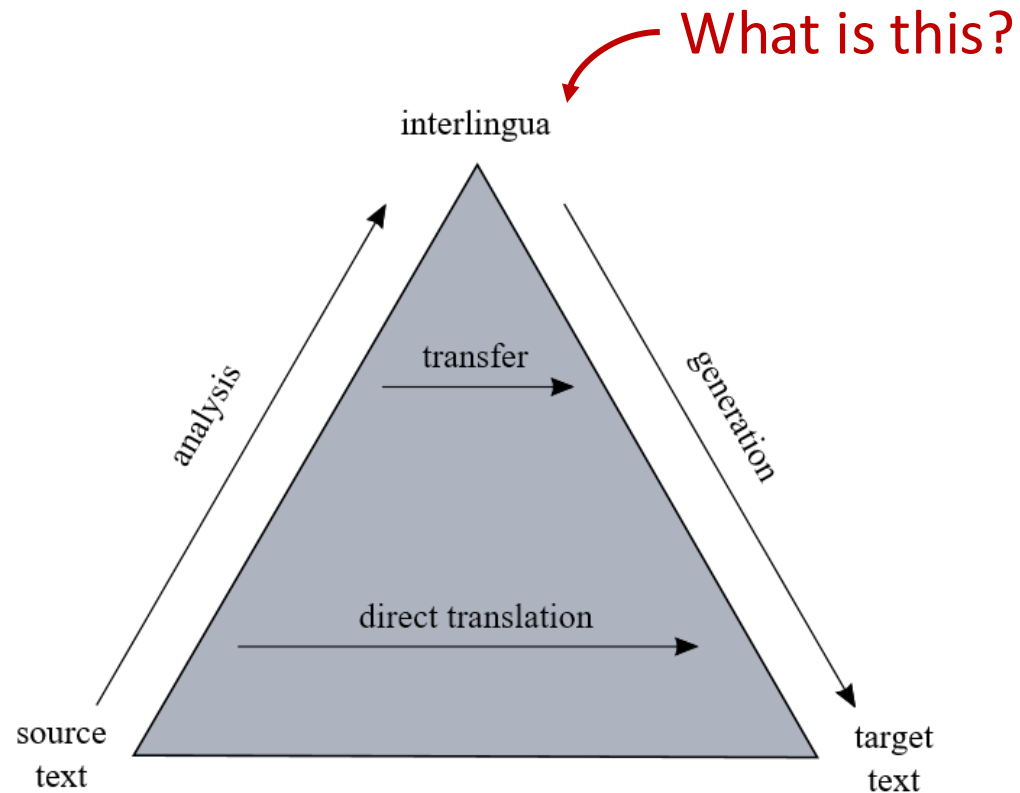
the cat sat on the table

η γάτα κάθισε στο τραπέζι

# Key Questions

- How to **represent** language?
- How to **compose** meaning?
- How do we translate from the “surface form” of language to a representation that a computer can process, and back?
  - How do we **encode** / **parse** the meaning of a sentence?
  - How do we **decode** / **generate** text from a meaning representation?
- What are the key methods / algorithms for **learning** these representations?

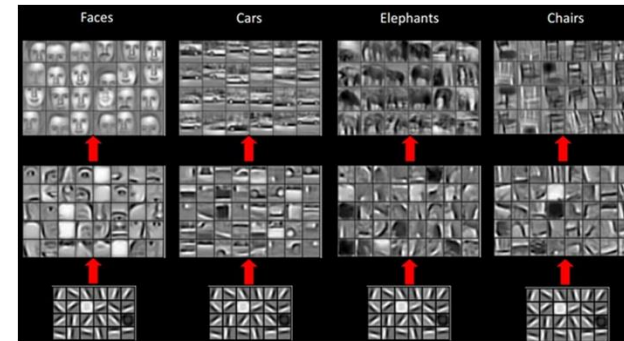
# What should the representation look like?



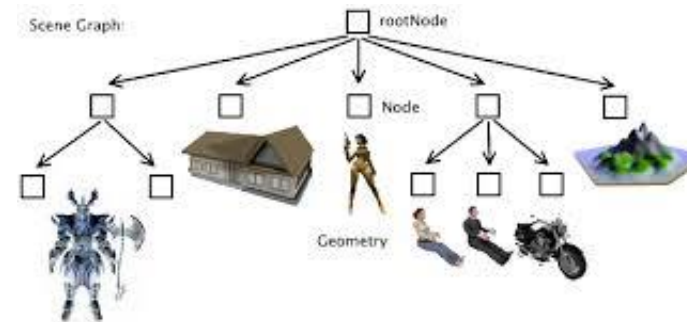
$$L = \begin{bmatrix} \vdots & \vdots & \vdots & \dots & \vdots & \vdots \\ \vdots & \vdots & \vdots & \dots & \vdots & \vdots \end{bmatrix}_n$$

the cat mat ...

$|V|$



Lambda expression:  
 $((\lambda x . (\lambda y . x)) (\lambda z . z)) (\lambda a . a)$



# Representations

How to represent the meaning of something?



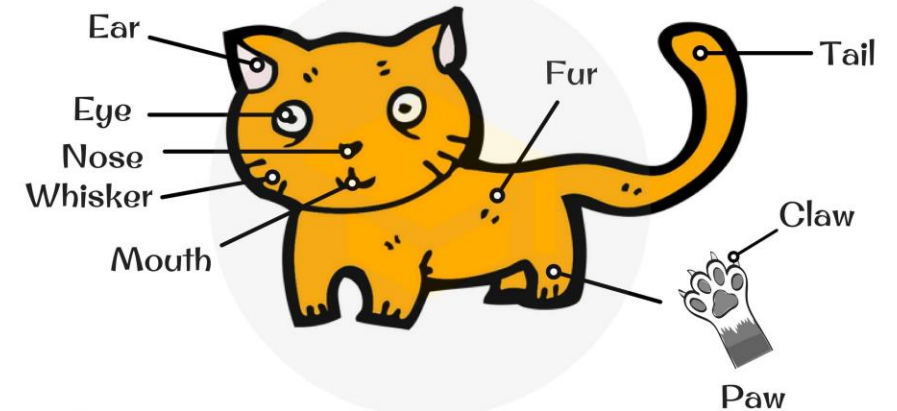
“cat”

**cat**: a small domesticated carnivore, *Felis domestica* or *F. catus*, bred in a number of varieties.

```
cat → {  
  isMammal: true  
  hasFur: true  
  hasLegs: true  
  meows: true  
  barks: false  
  height: 9.1 – 9.8 in  
  weight: 7.9 – 9.9 lbs  
  ...  
}
```

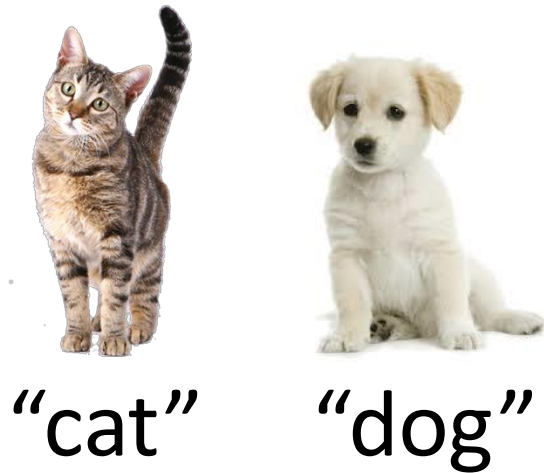
Attributed  
representation

## Parts of a cat



TESL.COM

# Representations



Representing meaning as vectors

- common representation space
- enables information sharing
- can be learned from data

- One-hot

cat = [0 0 0 1 0 0 0]

dog = [0 0 0 0 0 1 0]

- Embeddings

cat = [0.04 1.79 -1.79 1.07 0.48]

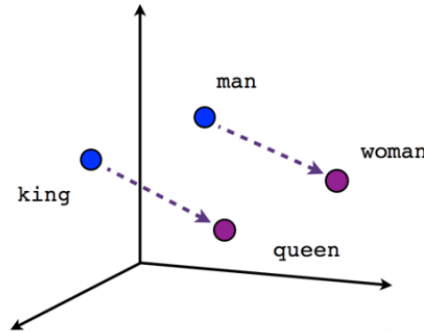
dog = [0.61 1.84 -1.12 0.52 0.53]



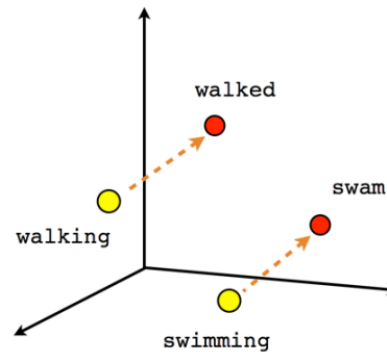
# Distributed representations

# Project words onto a continuous vector space

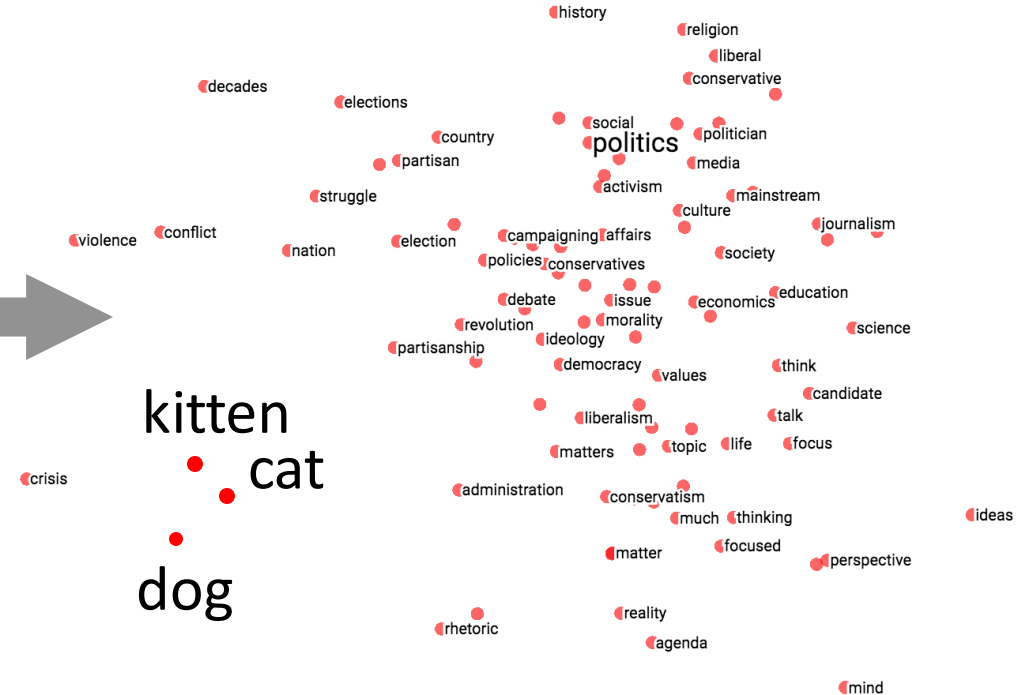
## Similar words closer to each other



Male-Female

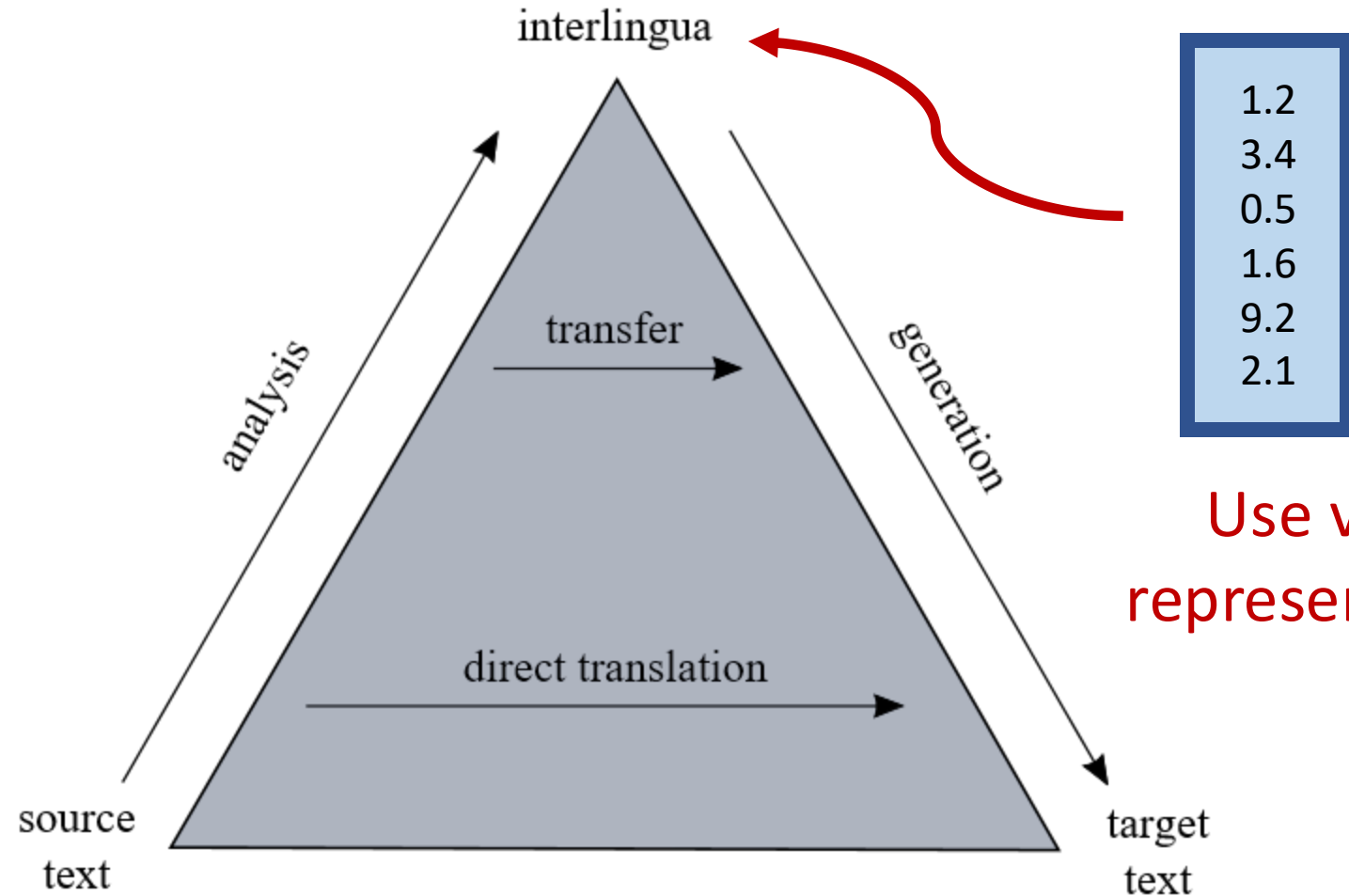


## Verb tense



$$v(\text{king}) - v(\text{man}) + v(\text{woman}) = v(\text{queen})$$

# Vauquois Triangle for translation

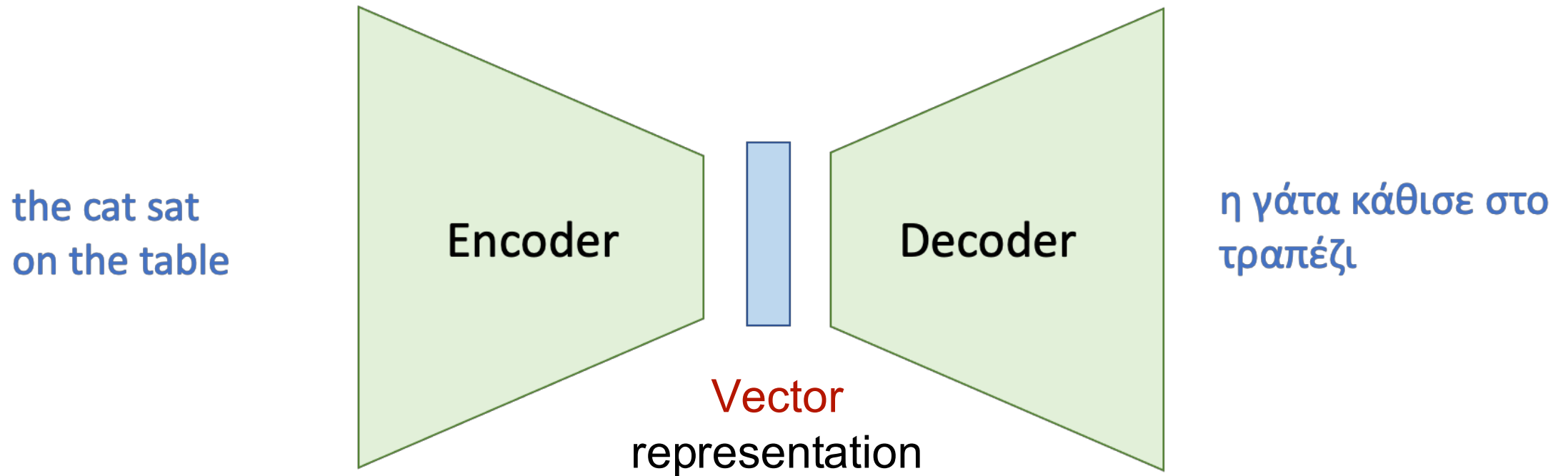


Use vector to  
represent meaning

the cat sat on the table

η γάτα κάθισε στο τραπέζι

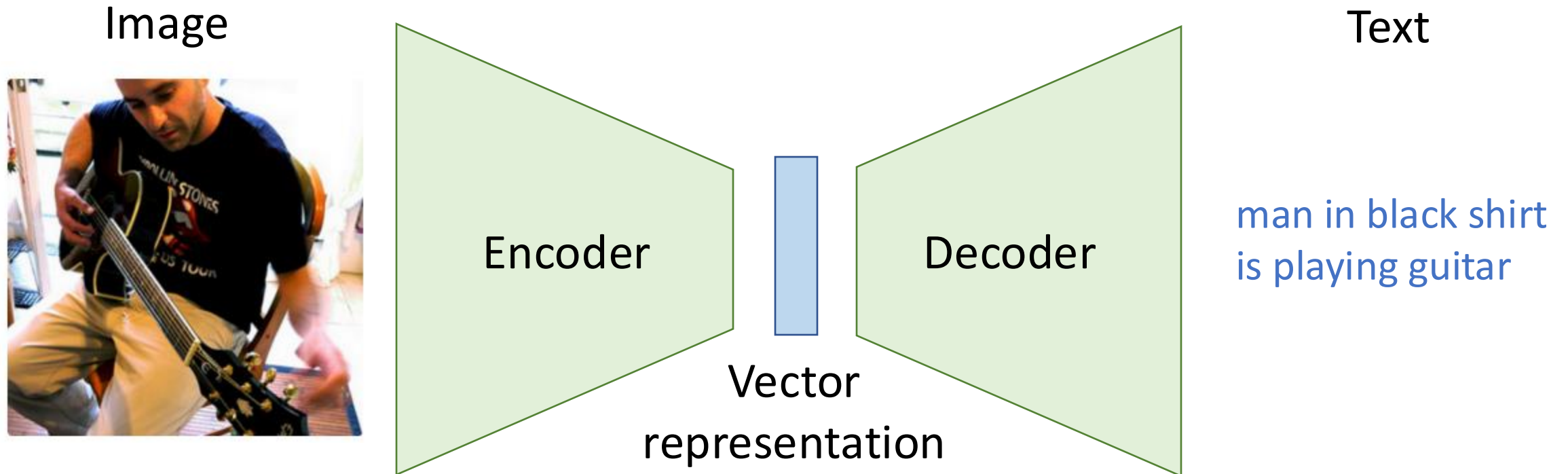
# Vector based representation



Common choice: flat unstructured vector of real numbers

- easy to work with
- well defined mathematical operations
- can be used to measure if two things are similar or not

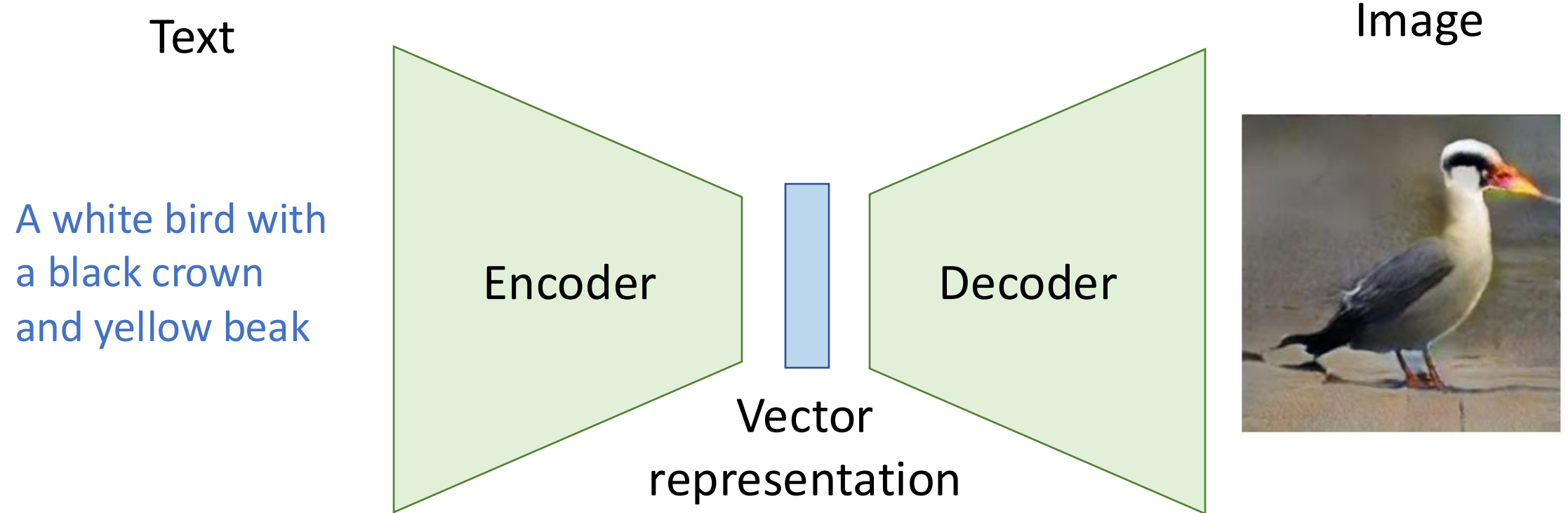
# Translating across modalities



## Image captioning

“Deep Visual-Semantic Alignments for Generating Image Descriptions”  
[Karpathy and Fei-Fei CVPR 2015]

# Translating across modalities



“StackGAN: Text to Photo-realistic Image Synthesis with Stacked Generative Adversarial Networks”  
[Zhang et al, ICCV 2017]

# Translating across modalities

Text

Brown colored dining table.  
It has four legs made of wood.

Encoder

Vector  
representation

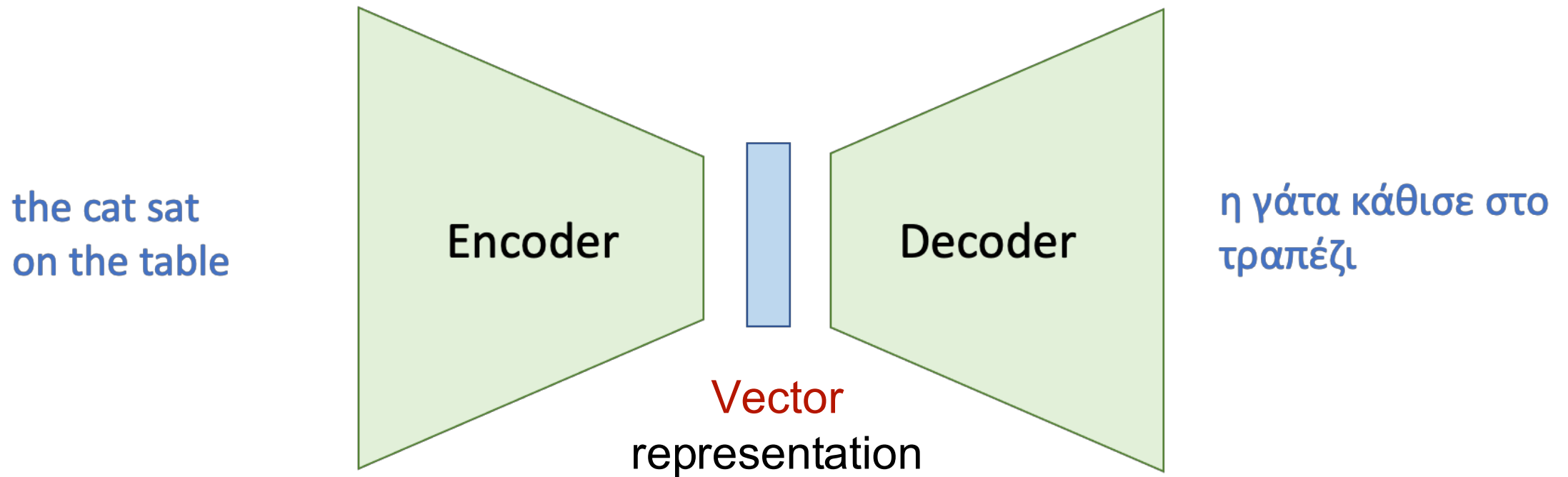
Decoder

3D Shape



“Text2Shape: Generating Shapes from Natural Language by Learning Joint Embeddings”  
[Chen et al, ACCV 2018]

# Vector based representation



Common choice: flat unstructured vector of real numbers

- easy to work with
- well defined mathematical operations
- can be used to measure if two things are similar or not



# How to encode language?

What is this function that takes the sentence and converts it into a vector?

the cat sat  
on the table

Encoder

Vector  
representation

Encoder

- How to convert the sentence into a meaning representation?

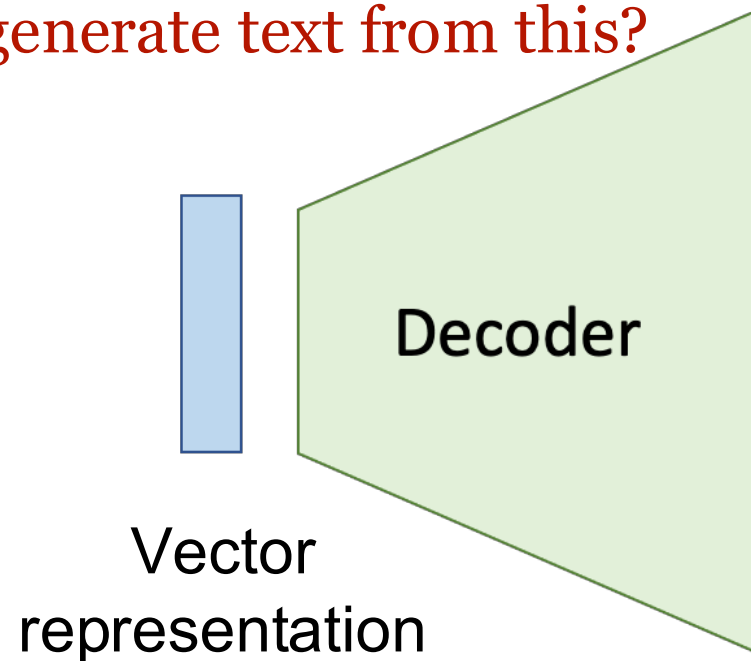
Understanding what is said  
(encoding, parsing, feature  
extraction)

# How to decode language?

How do we generate text from this?

Decoder

- How to take the vector representation and generate a sequence of words?



Deciding what to say  
(decoding, generating)

# How to encode language?

What is this function that takes the sentence and converts it into a vector?

the cat sat  
on the table

Encoder

Vector  
representation

Properties we want

- Be able to represent a single word
- Be able to represent the meaning of a sentence as a composition of the meaning of each word

Understanding what is said  
(encoding, parsing, feature  
extraction)

# How to learn representations for words?



"cat"



"dog"

start with representing words as discrete tokens

- One-hot

cat = [0 0 0 1 0 0 0]

dog = [0 0 0 0 0 1 0]

learn about neural models that use vectors

- Embeddings

cat = [0.04 1.79 -1.79 1.07 0.48]

dog = [0.61 1.84 -1.12 0.52 0.53]

use **context** to represent words

...government debt problems turning into **banking** crises as happened in 2009...  
...saying that Europe needs unified **banking** regulation to replace the hodgepodge...  
...India has just given its **banking** system a shot in the arm...

These **context words** will represent **banking**

learn a dense vector representation that can be used to **predict** words from their context

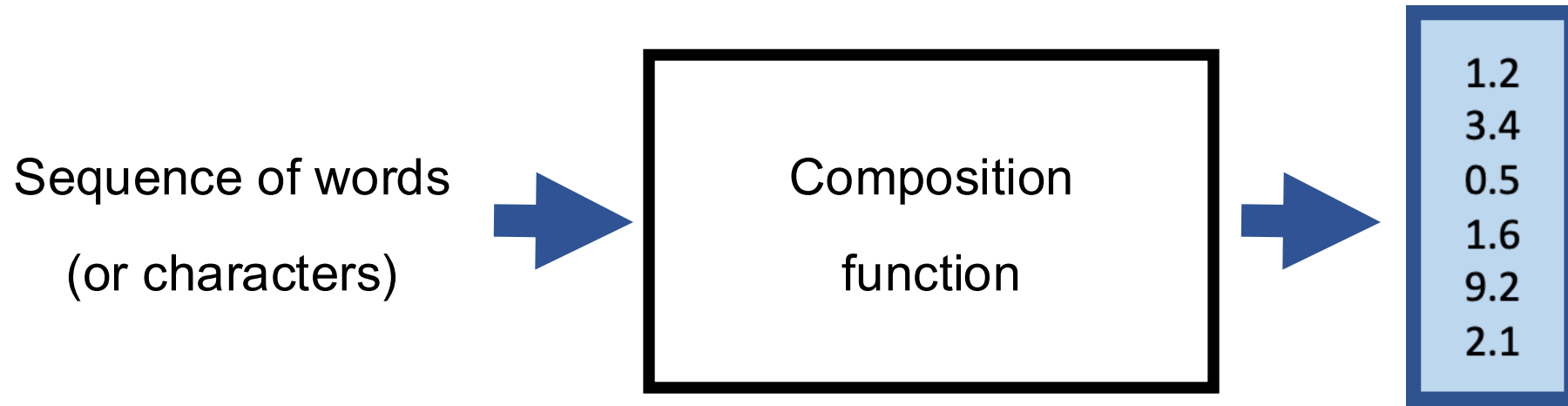
- Predict what goes next

The cat sat on the \_\_\_\_\_

- Fill in the blank

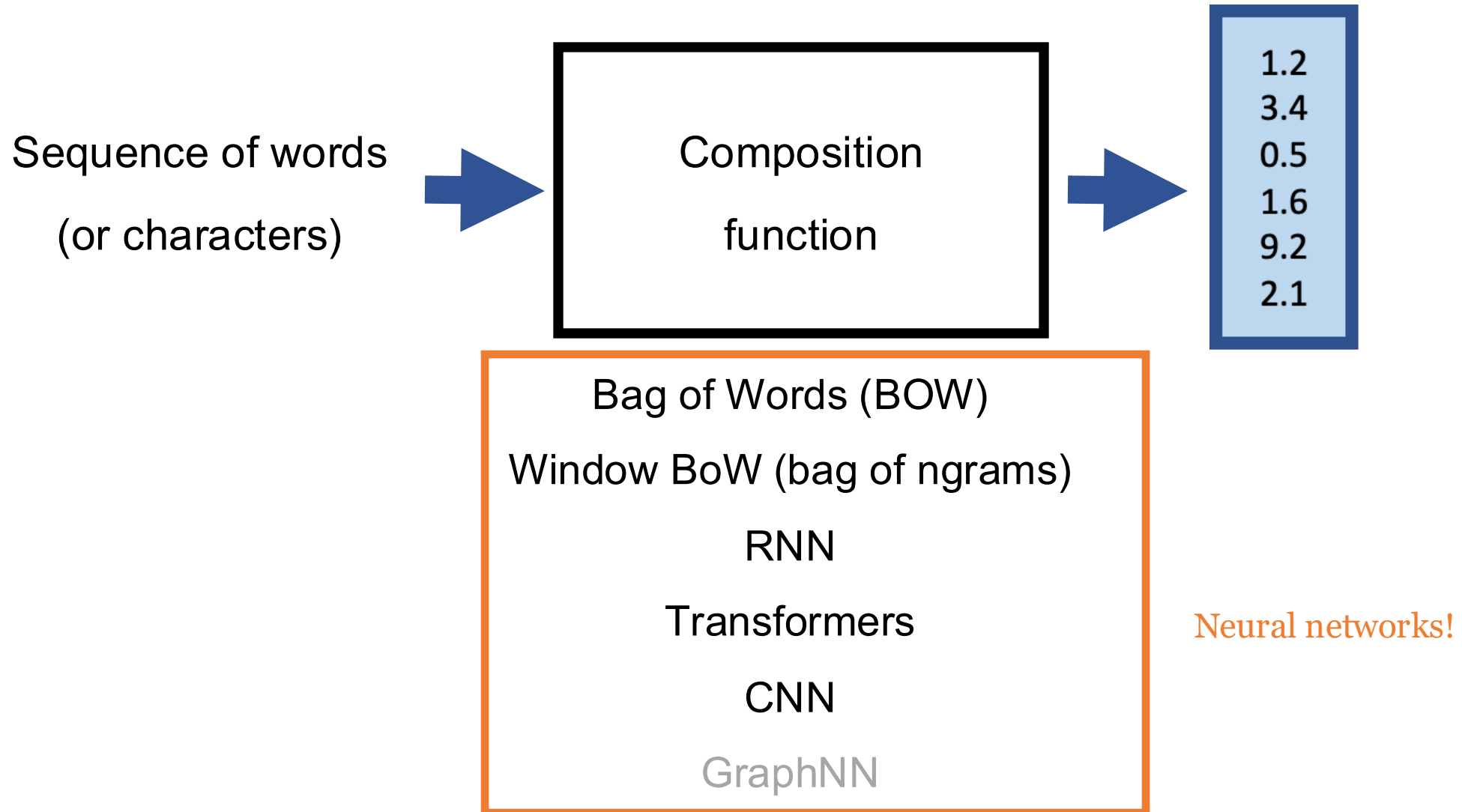
A bottle of \_\_\_\_\_ is on the table.

# How to **compose** language?

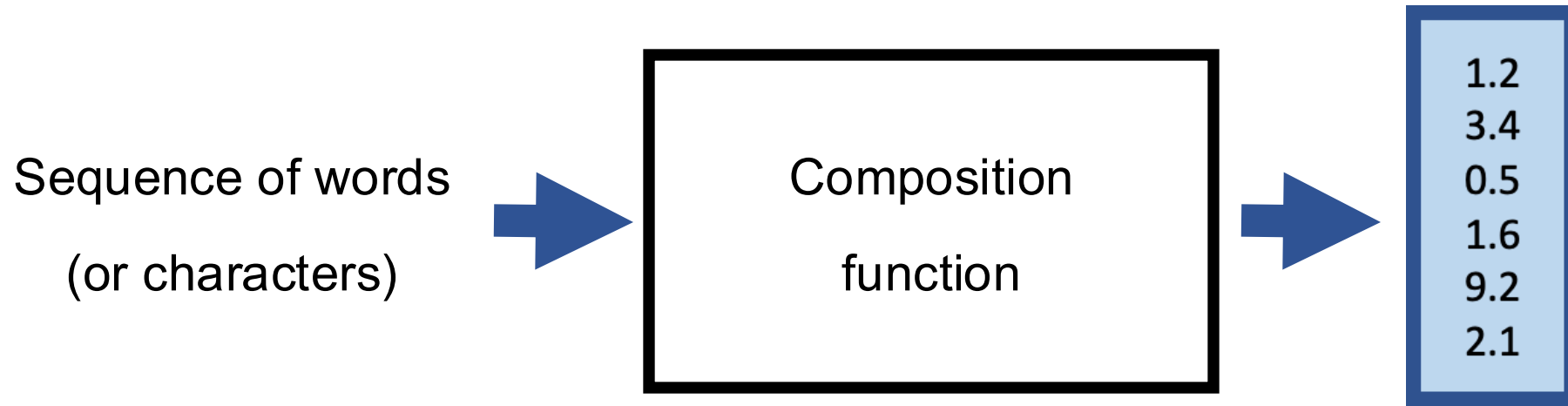


Bag of Words (BOW)  
Window BoW (bag of ngrams)  
RNN  
Transformers  
CNN  
GraphNN

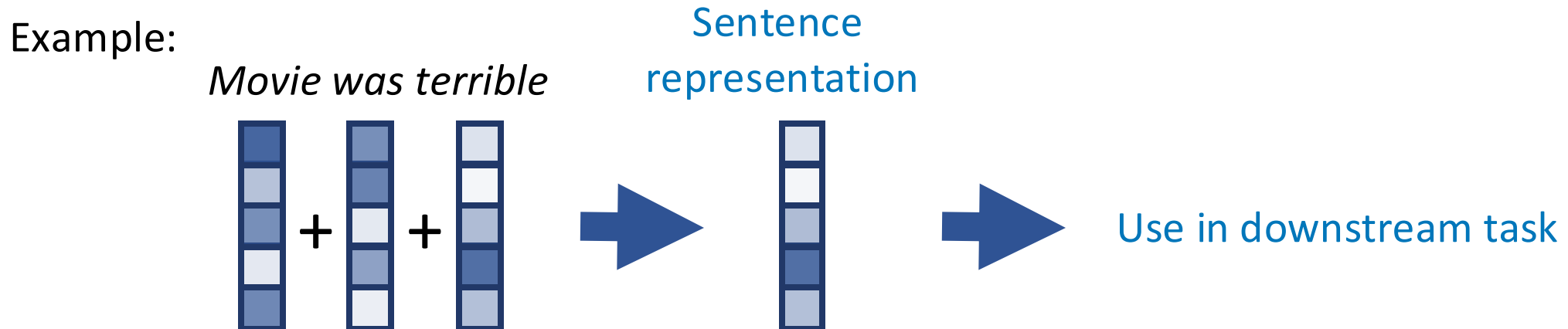
# How to **compose** language?



# How to **compose** language?



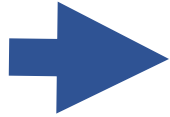
Simplest: take weighted average





# How to use the learned representation?

What do we do with this vector representation after we get it?



Use in downstream task

Vector  
representation

Do prediction: classify as spam/not spam

Use to generate text (also known as **decoding**)

Can also be used to generate other stuff (like images!)

# Thank you!



Getty Images

<https://www.thoughtco.com/syntactic-ambiguity-grammar-1692179>

